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(54) **BUFFER DESIGN FOR PACKAGE INTEGRATION**

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H01L 21/48 (2006.01)
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(52) **U.S. Cl.**

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(Continued)

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H01L 25/50; **H01L 2225/1035-1058**
See application file for complete search history.

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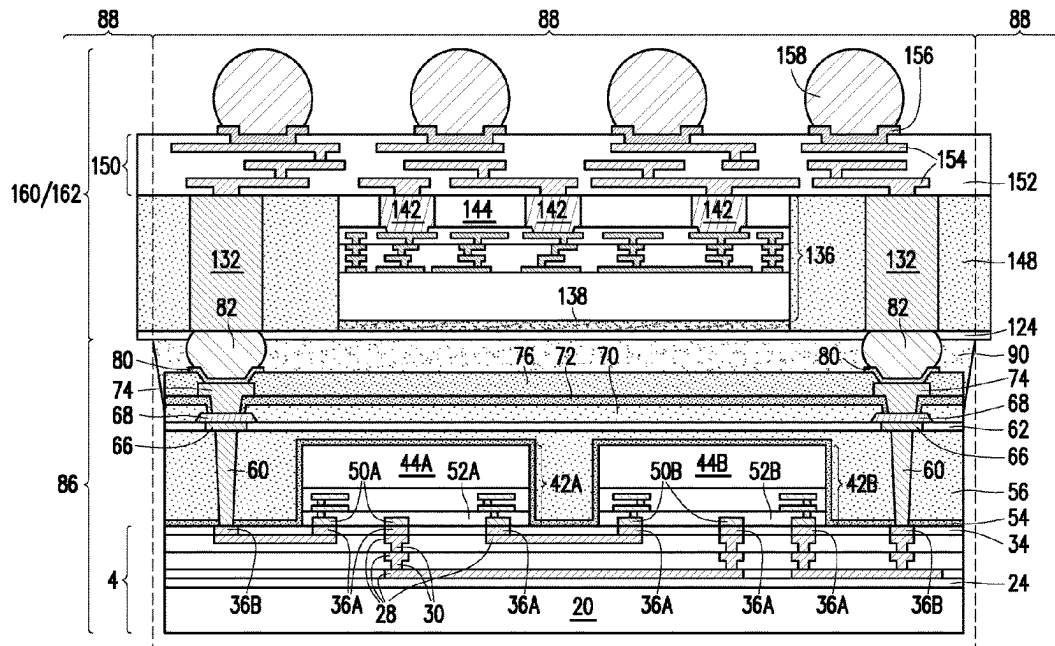
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(57) **ABSTRACT**

A method of forming a package includes bonding a device die to an interposer wafer, with the interposer wafer including metal lines and vias, forming a dielectric region to encircle the device die, and forming a through-via to penetrate through the dielectric region. The through-via is electrically connected to the device die through the metal lines and the vias in the interposer wafer. The method further includes forming a polymer layer over the dielectric region, and forming an electrical connector. The electrical connector is electrically coupled to the through-via through a conductive feature in the polymer layer. The interposer wafer is sawed to separate the package from other packages.

20 Claims, 22 Drawing Sheets



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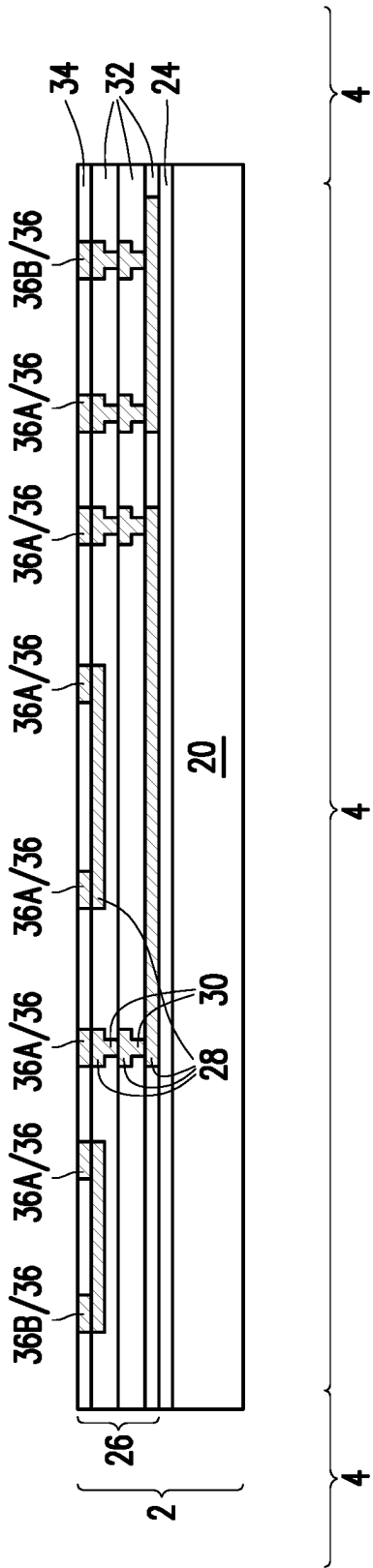
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H01L 25/065 (2023.01)
H01L 25/10 (2006.01)
H01L 25/18 (2023.01)

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(2013.01); *H01L 25/105* (2013.01); *H01L*
25/18 (2013.01); *H01L 2225/1035* (2013.01);
H01L 2225/1041 (2013.01); *H01L 2225/1058*
(2013.01)

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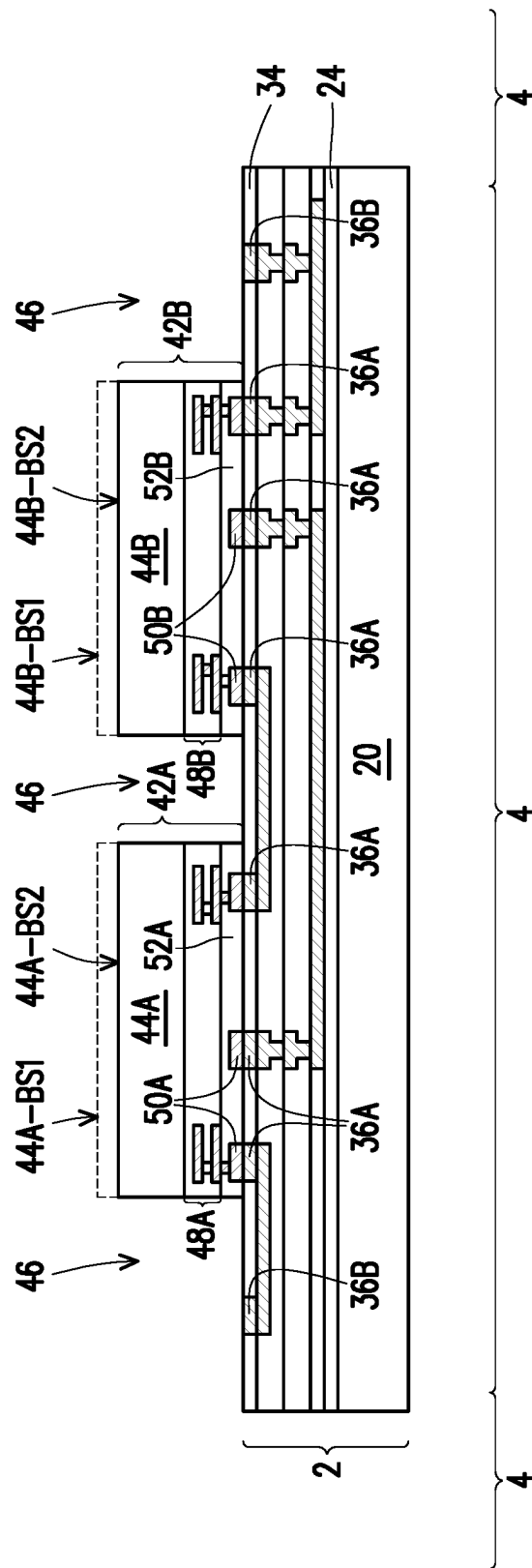


FIG. 2

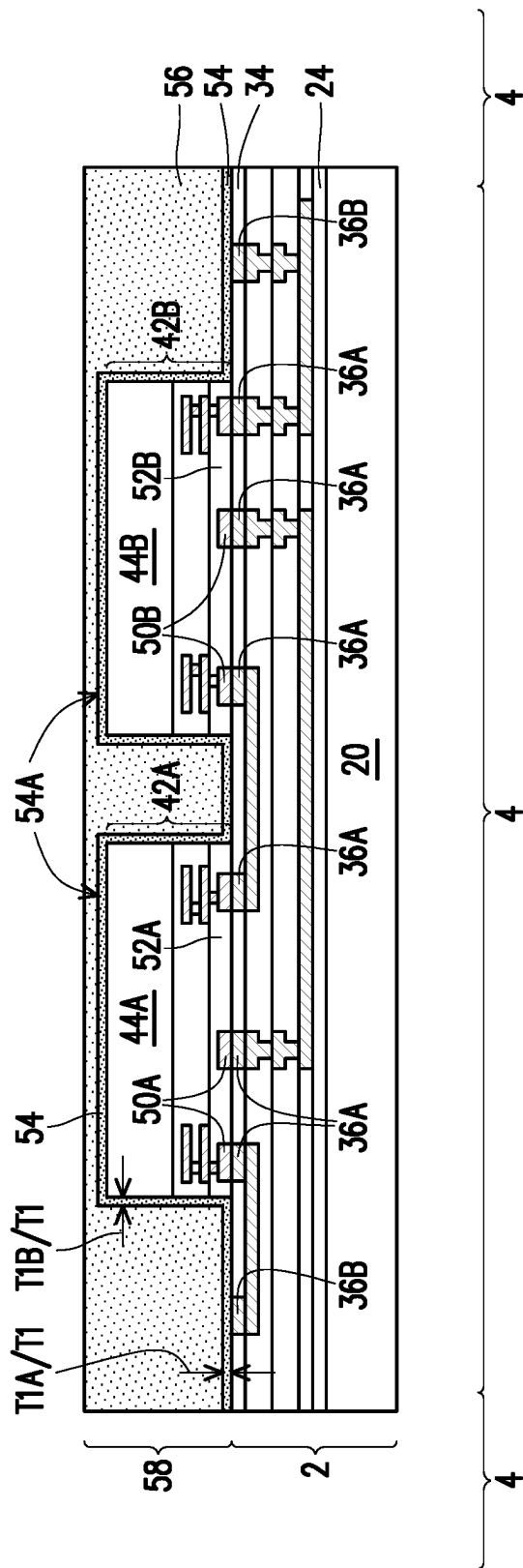


FIG. 3

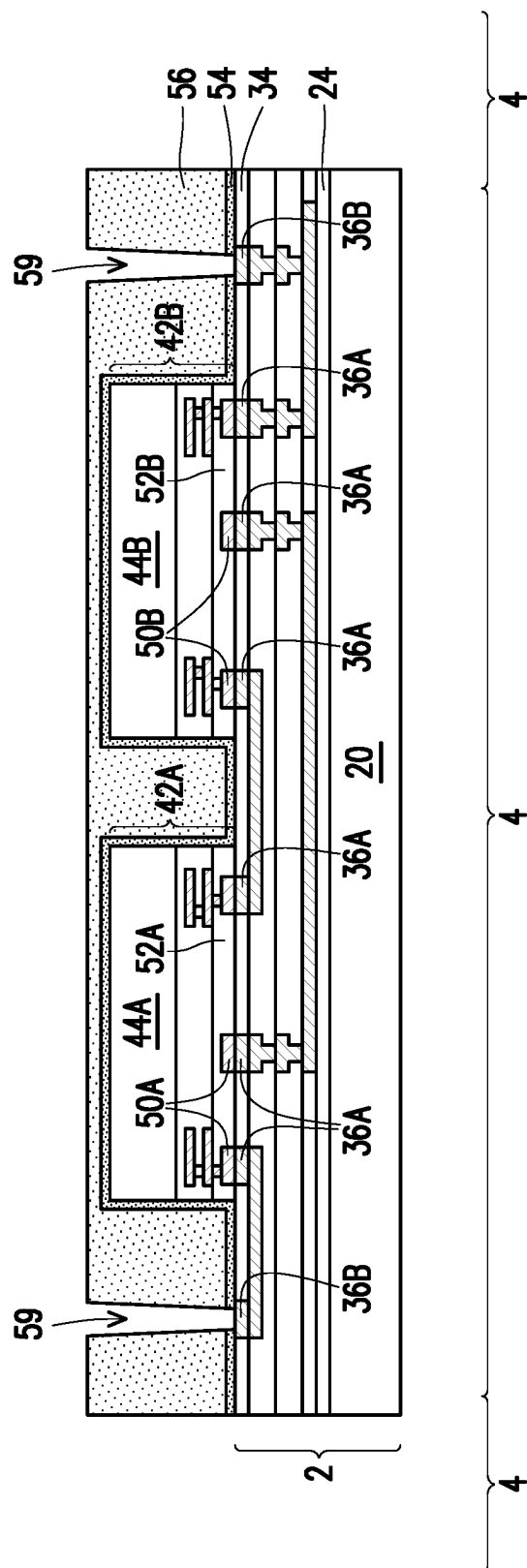


FIG. 4

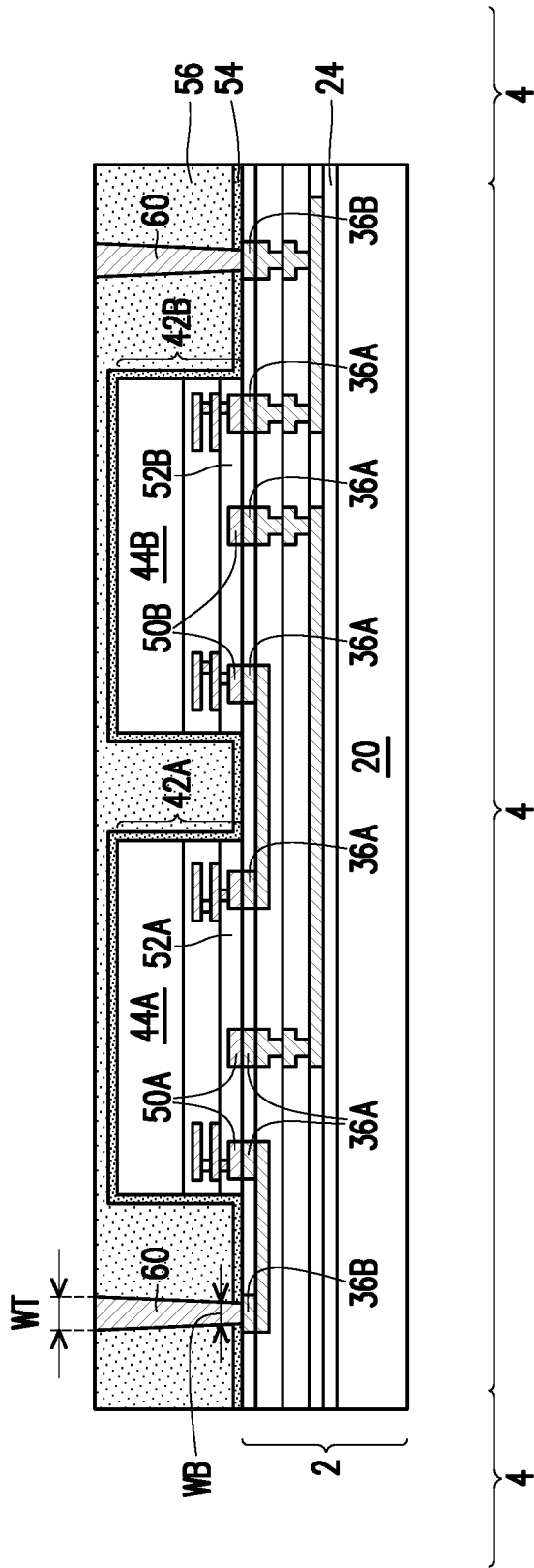


FIG. 5

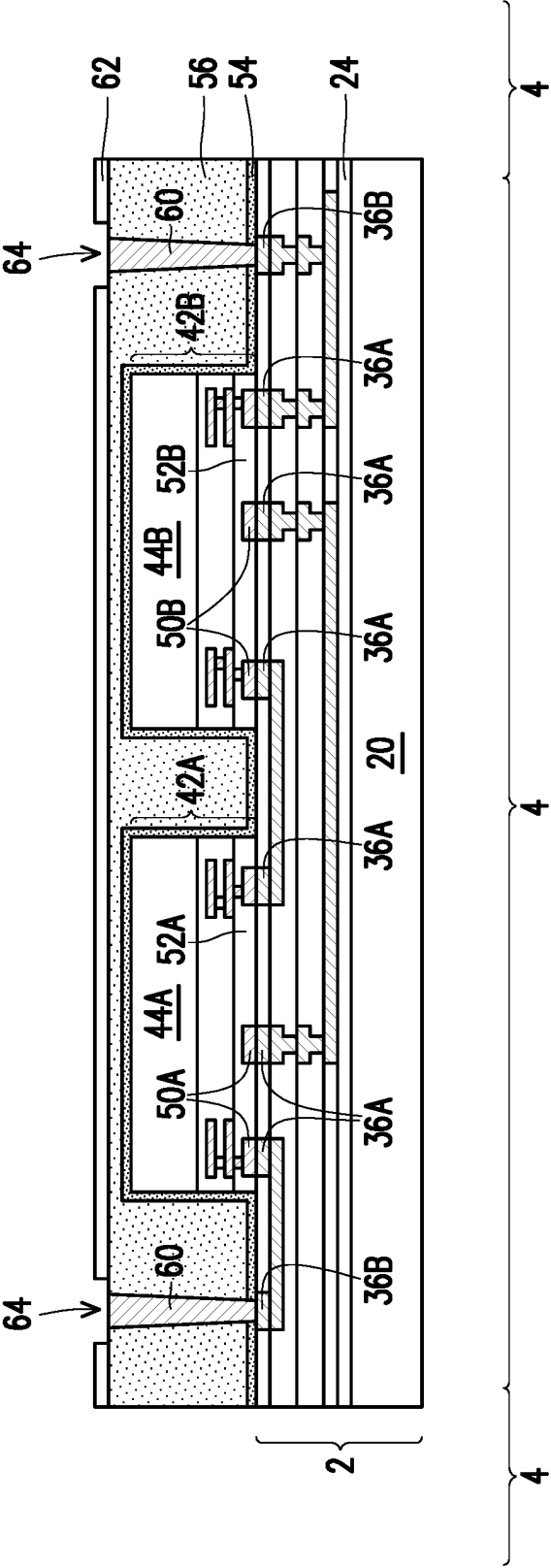


FIG. 6

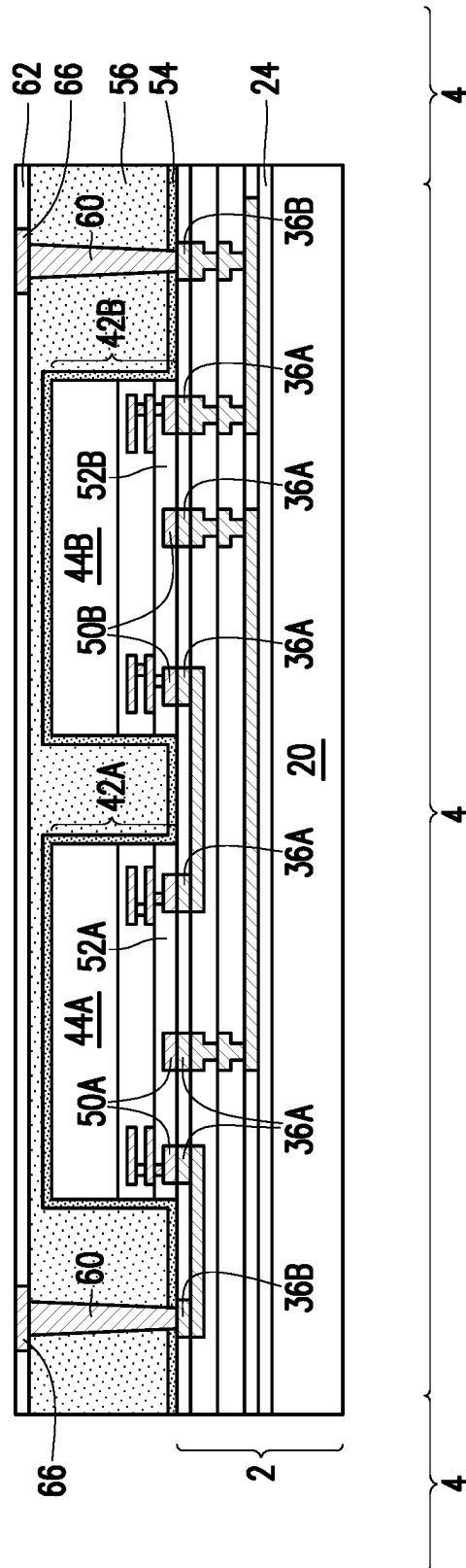


FIG. 7

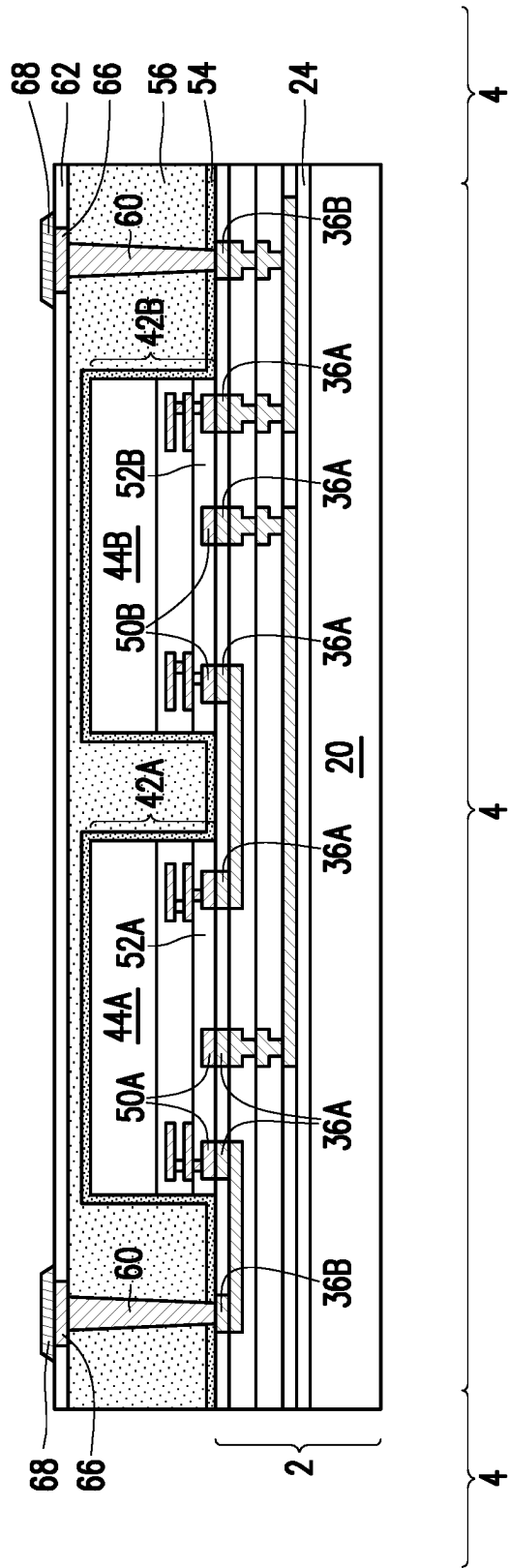


FIG. 8

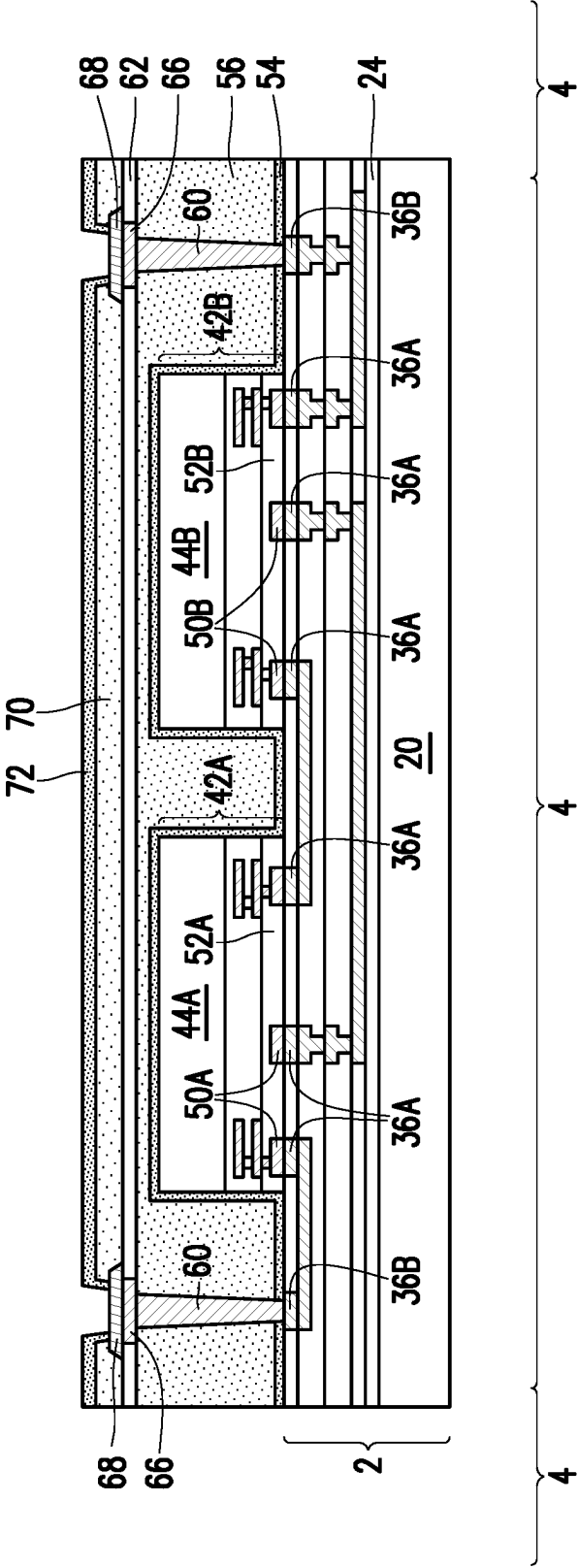


FIG. 9

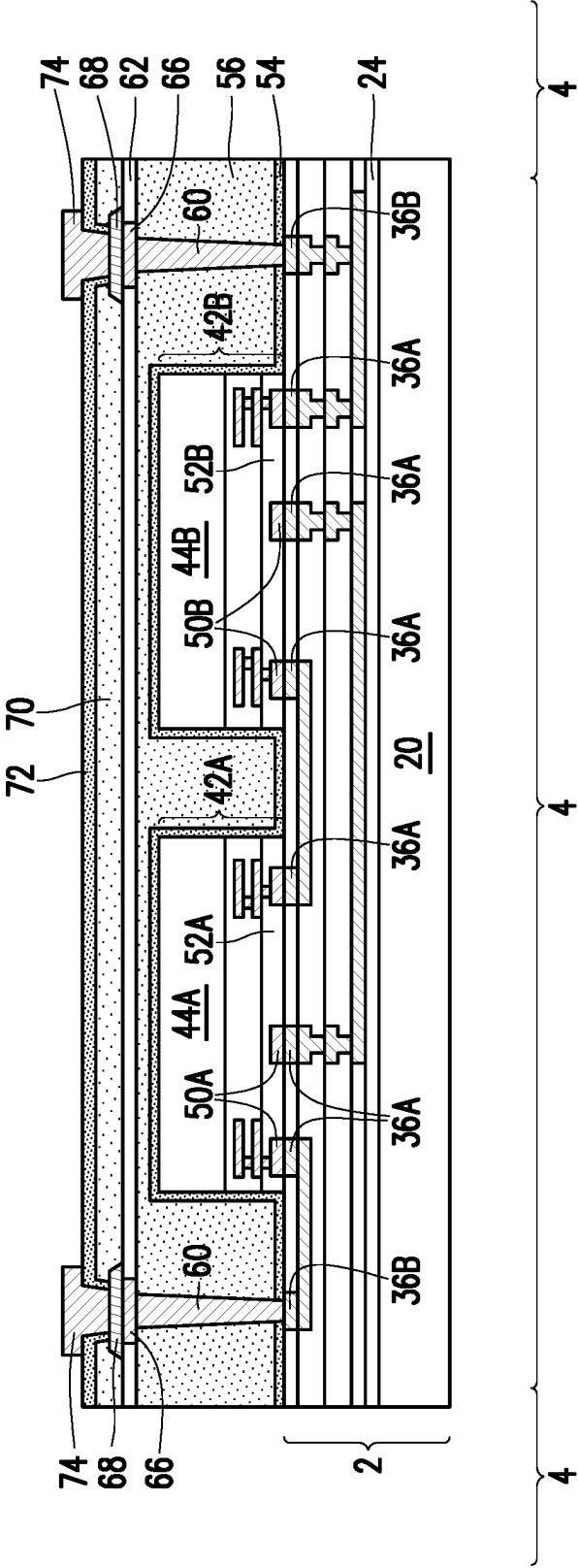


FIG. 10

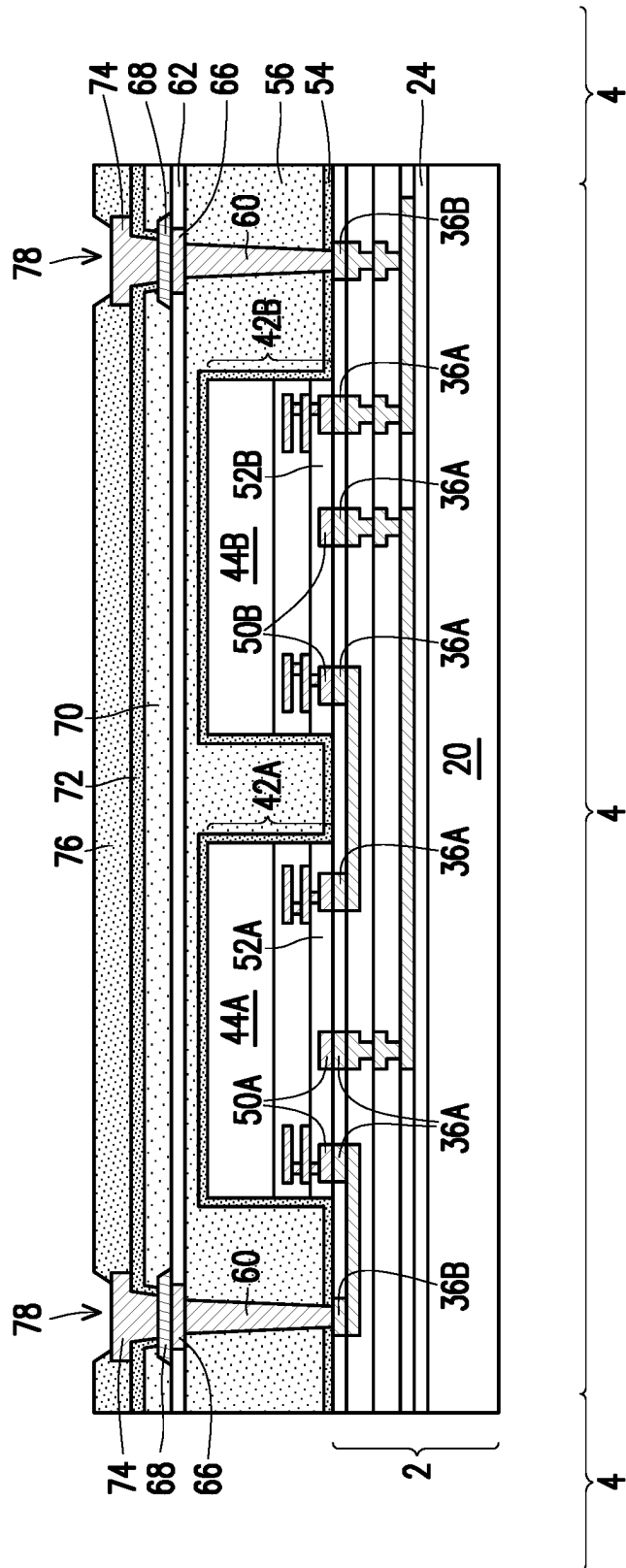


FIG. 11

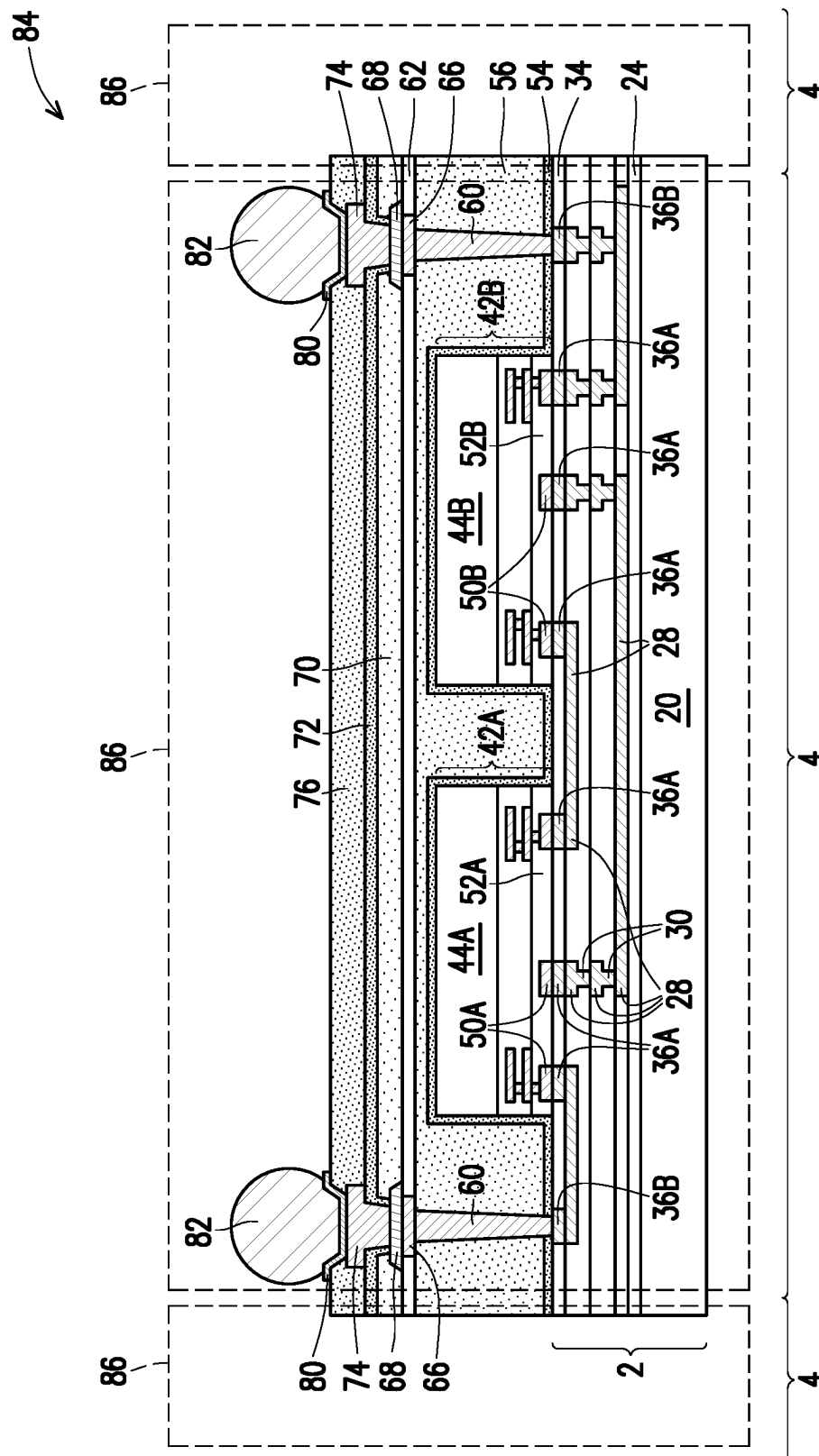


FIG. 12

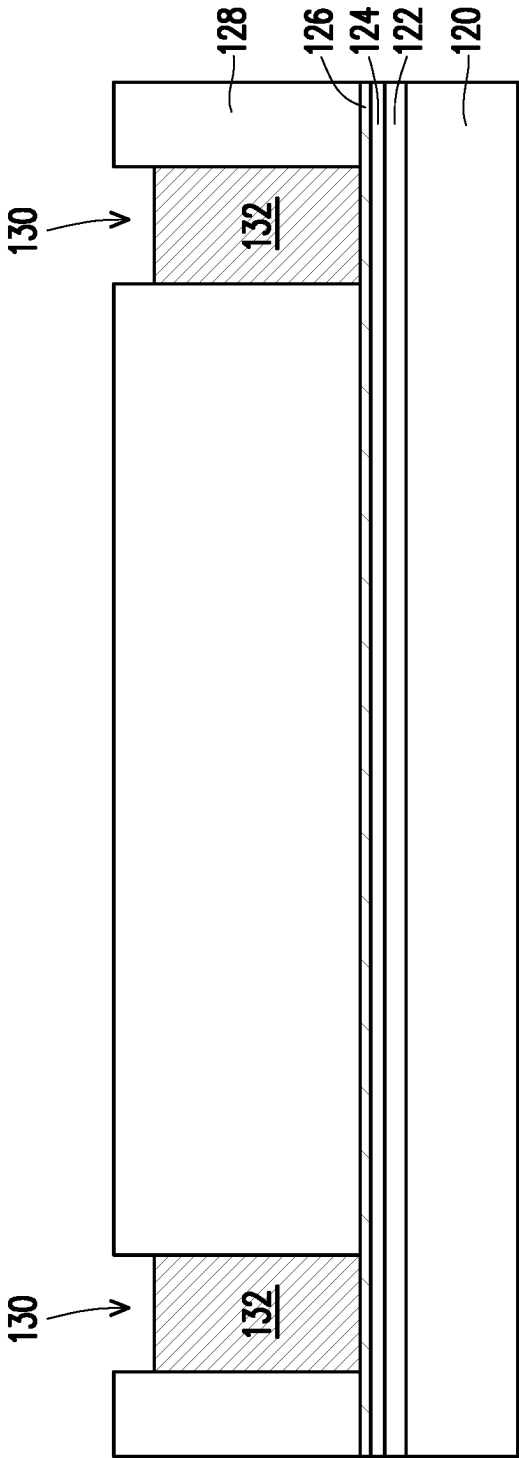


FIG. 13

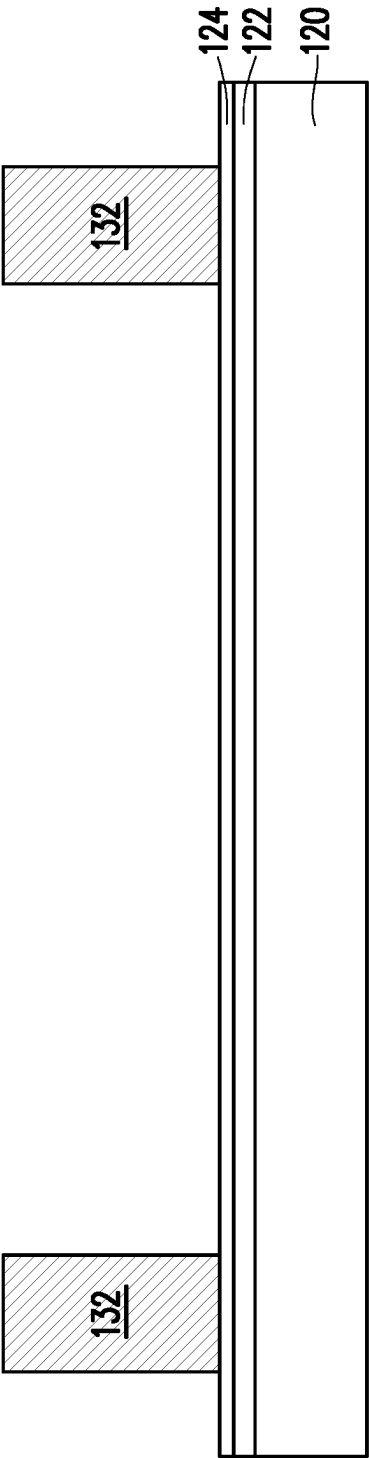


FIG. 14

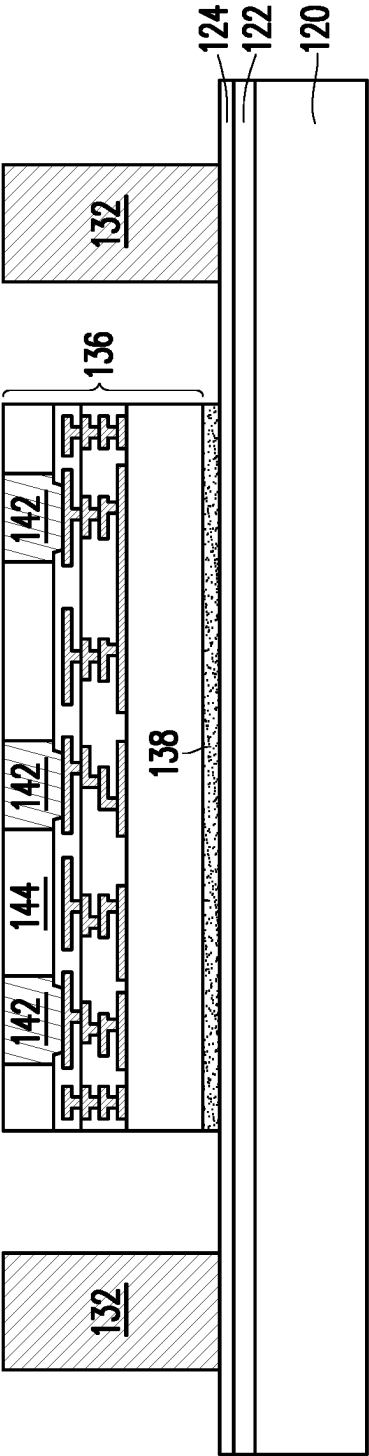


FIG. 15

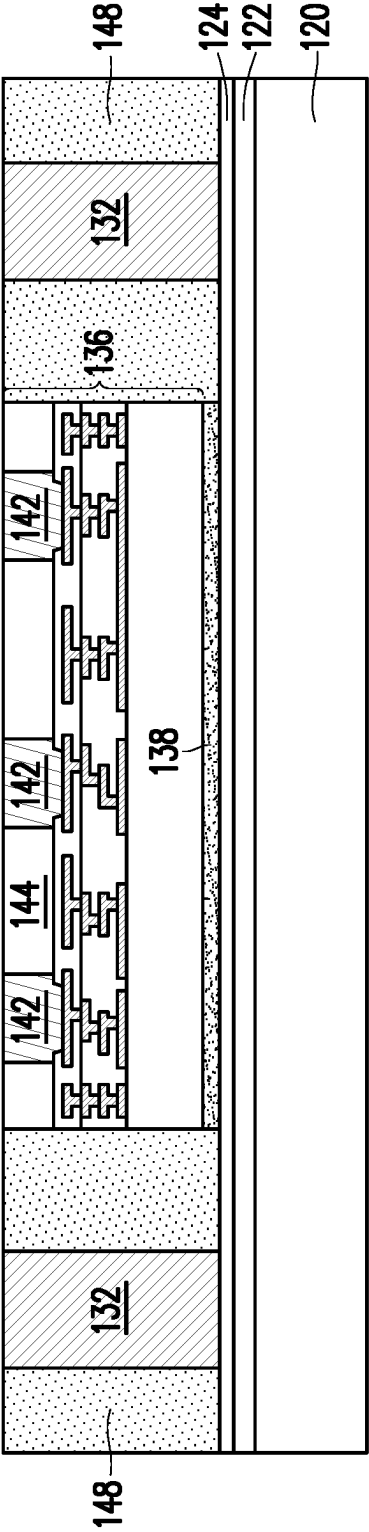


FIG. 16

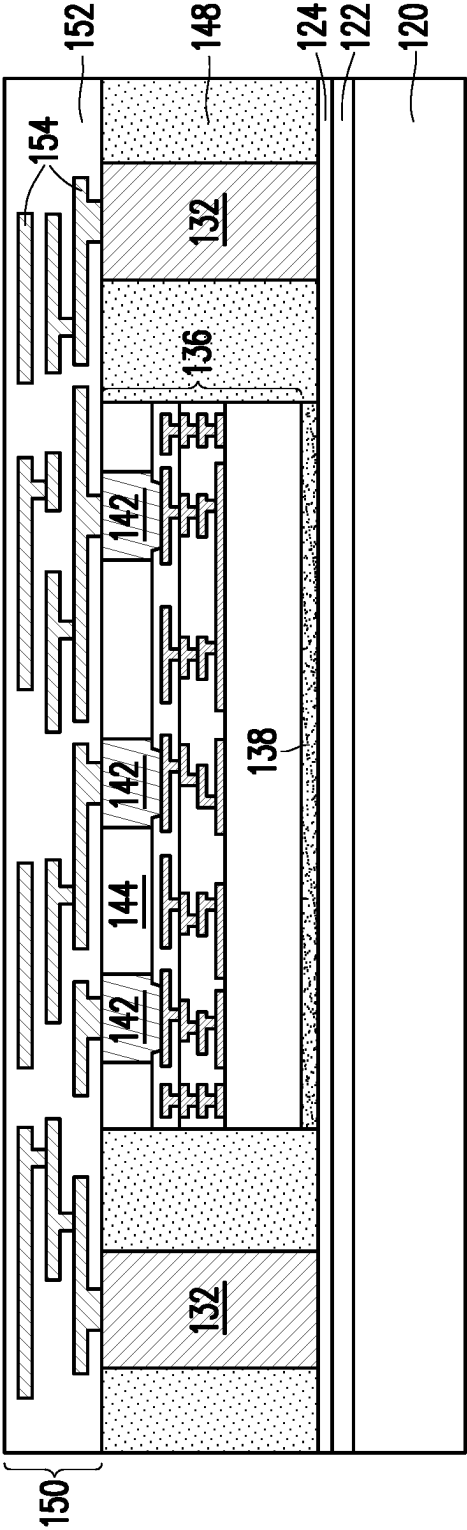


FIG. 17

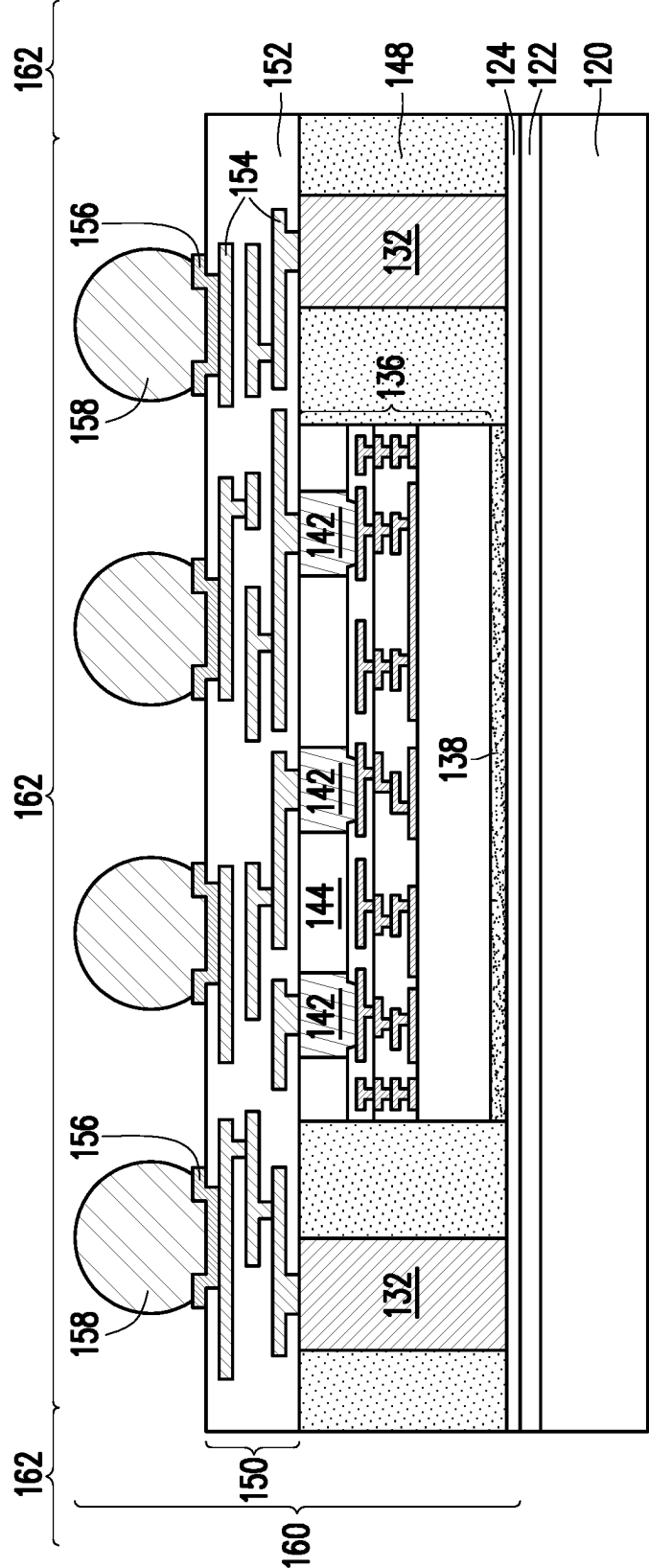


FIG. 18

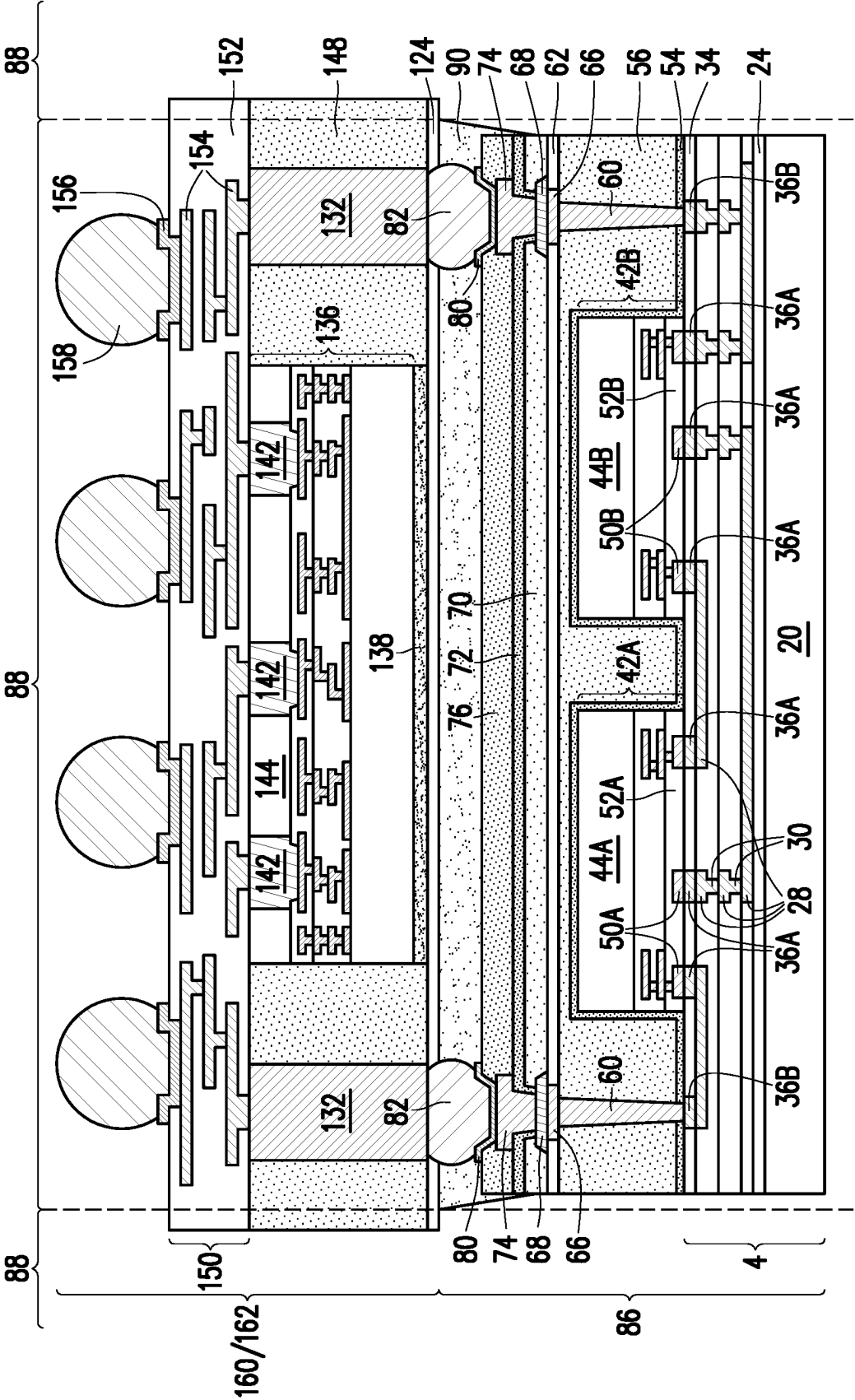


FIG. 19

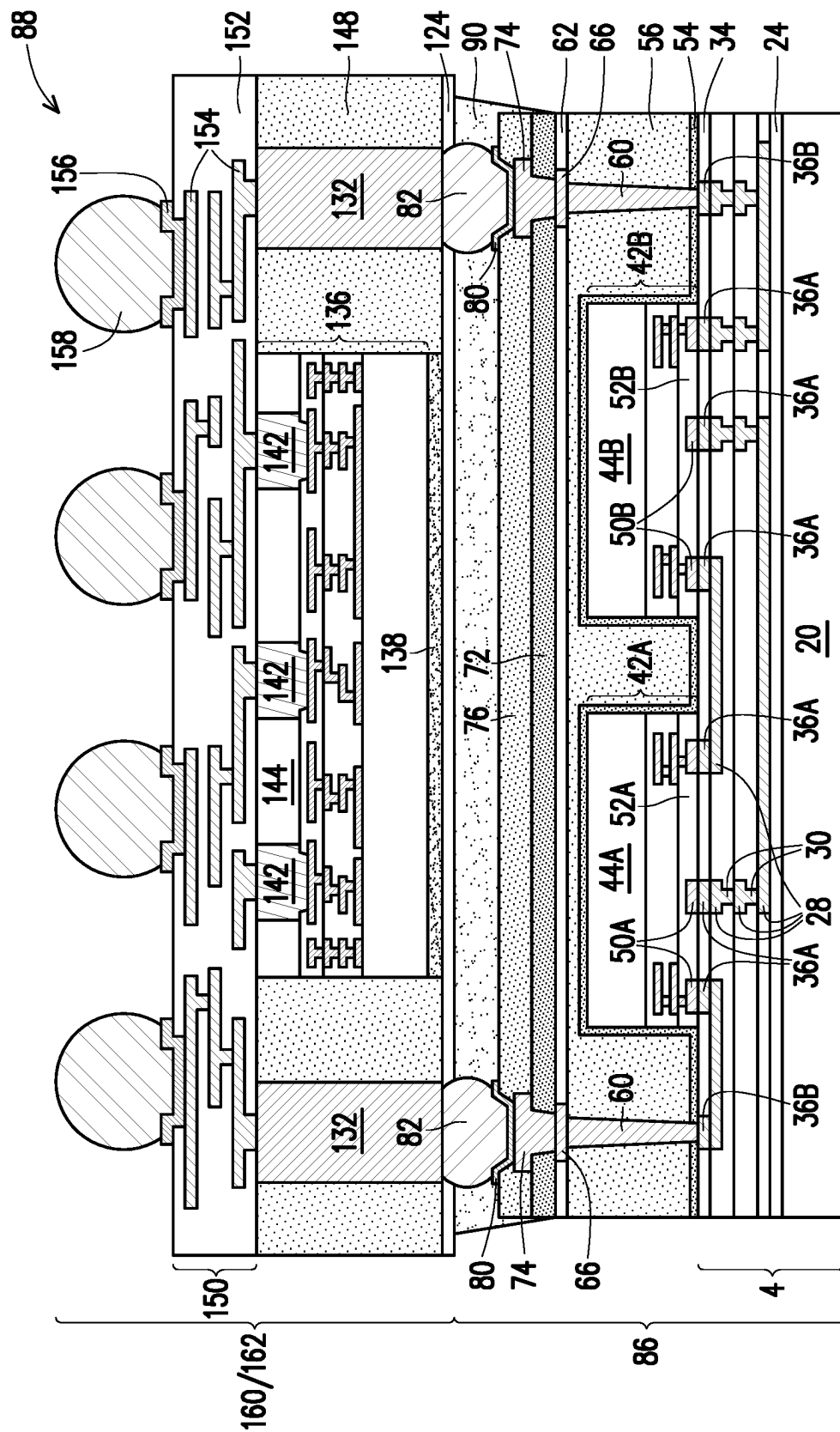


FIG. 20

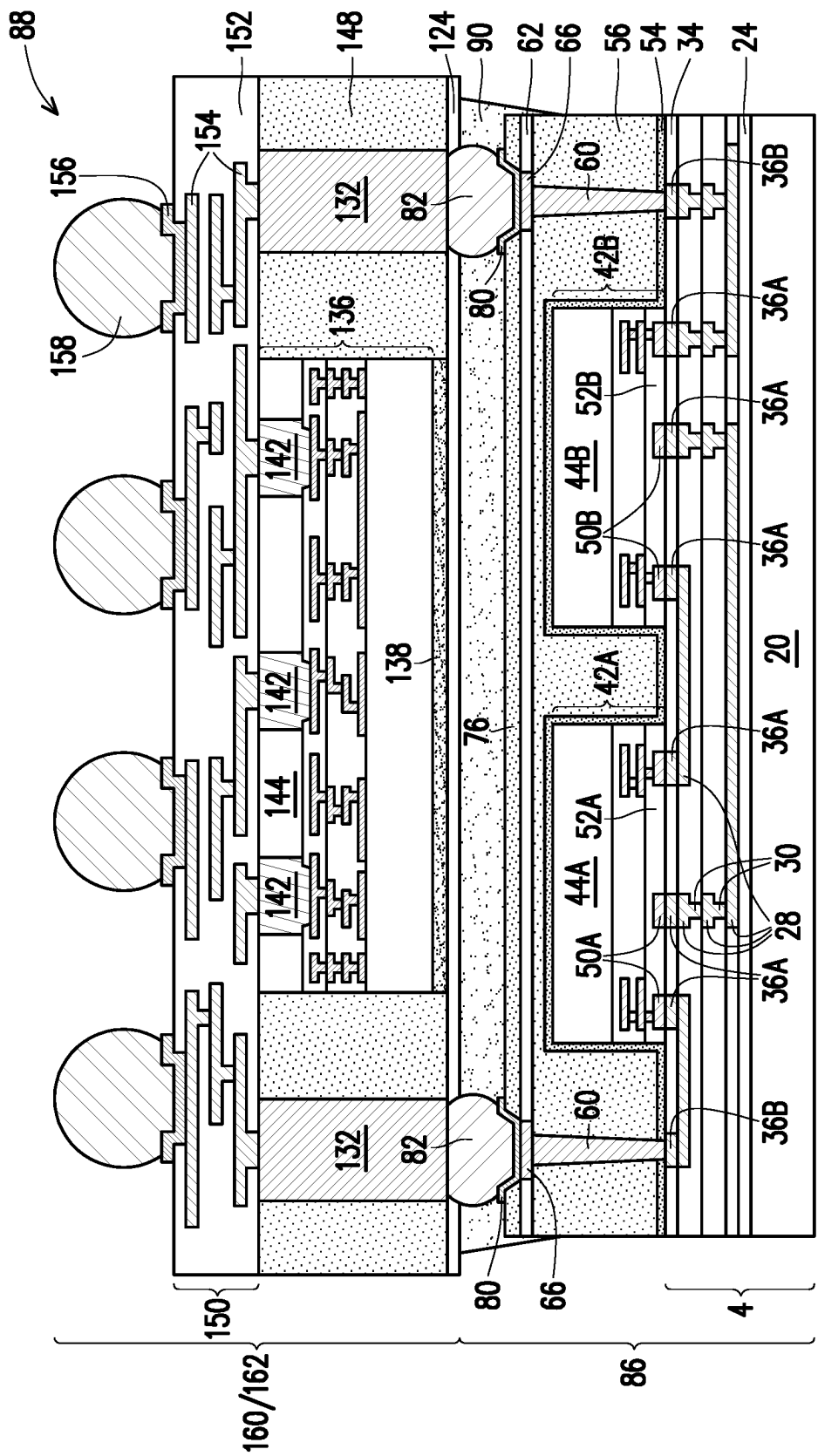


FIG. 21

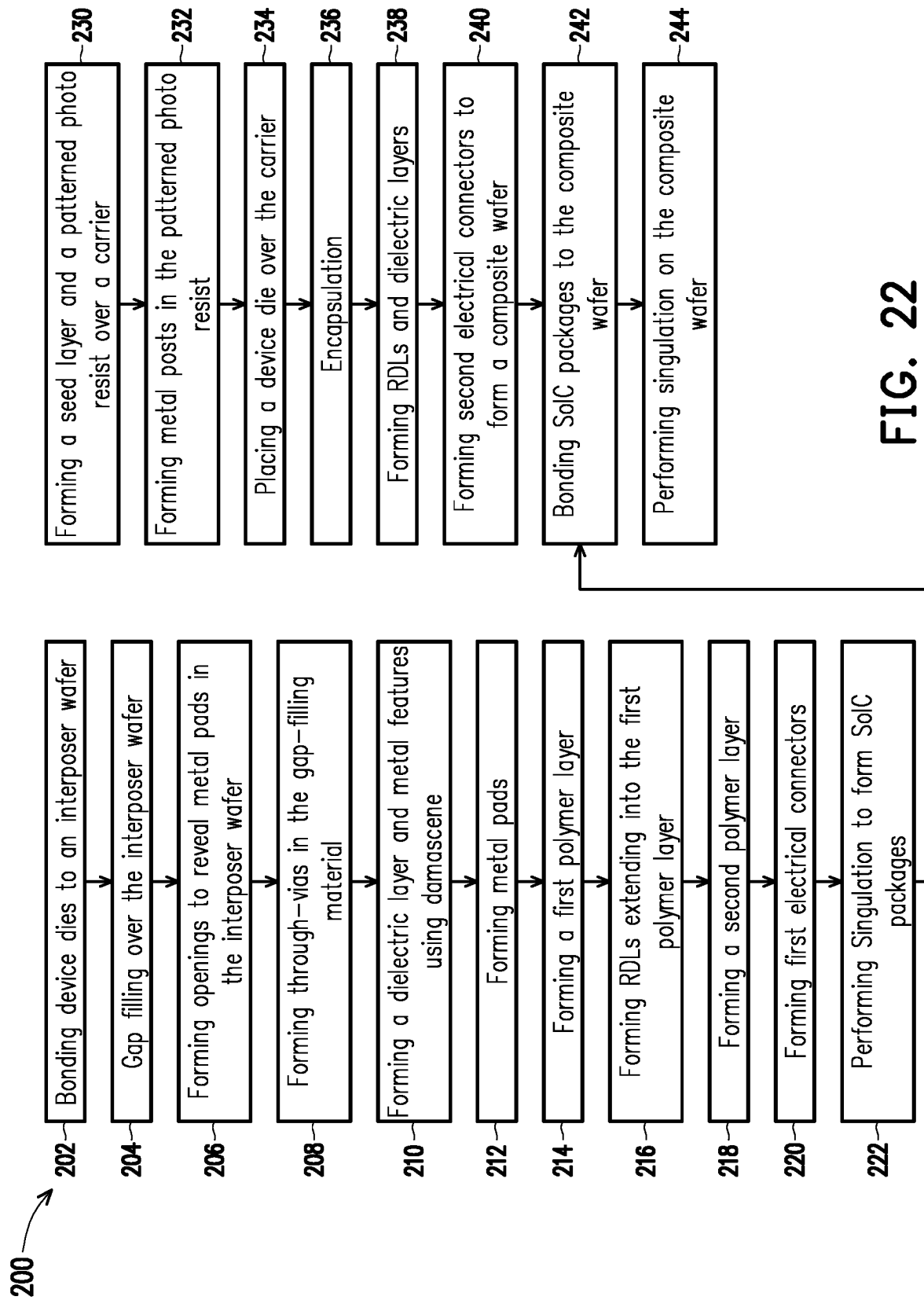


FIG. 22

BUFFER DESIGN FOR PACKAGE INTEGRATION

PRIORITY CLAIM AND CROSS-REFERENCE

This application is a continuation of U.S. patent application Ser. No. 17/181,720, entitled "Buffer Design for Package Integration," filed Feb. 22, 2021, which is a divisional of U.S. patent application Ser. No. 16/120,752, entitled "Buffer Design for Package Integration," filed Sep. 4, 2018, now U.S. Pat. No. 10,930,633, issued Feb. 23, 2021, which claims the benefit of the U.S. Provisional Application No. 62/691,989, filed Jun. 29, 2018, and entitled "Buffer Design for INFO Package System Integration," which applications are hereby incorporated herein by reference.

BACKGROUND

With the evolving of semiconductor technologies, semiconductor chips/dies are becoming increasingly smaller. In the meantime, more functions need to be integrated into the semiconductor dies. Accordingly, the semiconductor dies need to have increasingly greater numbers of I/O pads packed into smaller areas, and the density of the I/O pads rises over time. As a result, the packaging of the semiconductor dies becomes more difficult, which adversely affects the yield of the packaging.

Conventional package technologies can be divided into two categories. In the first category, dies on a wafer are packaged before they are sawed. This packaging technology has some advantageous features, such as a greater throughput and a lower cost. Further, less underfill or molding compound is needed. However, this packaging technology also suffers from drawbacks. Since the sizes of the dies are becoming increasingly smaller, and the respective packages can only be fan-in type packages, in which the I/O pads of each die are limited to a region directly over the surface of the respective die. With the limited areas of the dies, the number of the I/O pads is limited due to the limitation of the pitch of the I/O pads. If the pitch of the pads is to be decreased, solder bridges may occur. Additionally, under the fixed ball-size requirement, solder balls must have a certain size, which in turn limits the number of solder balls that can be packed on the surface of a die.

In the other category of packaging, dies are sawed from wafers before they are packaged. An advantageous feature of this packaging technology is the possibility of forming fan-out packages, which means the I/O pads on a die can be redistributed to a greater area than the die, and hence the number of I/O pads packed on the surfaces of the dies can be increased. Another advantageous feature of this packaging technology is that "known-good-dies" are packaged, and defective dies are discarded, and hence cost and effort are not wasted on the defective dies.

BRIEF DESCRIPTION OF THE DRAWINGS

Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

FIGS. 1 through 12 are cross-sectional views of intermediate stages in the formation of a System on Integrate Chip (SoIC) package in accordance with some embodiments.

FIGS. 13 through 18 are cross-sectional views of intermediate stages in the formation of an Integrated Fan-Out (InFO) package in accordance with some embodiments.

FIG. 19 illustrates a cross-sectional view of a package including a SoIC package bonded to an InFO package in accordance with some embodiments.

FIGS. 20 and 21 illustrate cross-sectional views of packages including SoIC packages bonded with InFO packages in accordance with some embodiments.

FIG. 22 illustrates a process flow for forming an integrated package including a SoIC package bonded to an InFO package in accordance with some embodiments.

DETAILED DESCRIPTION

The following disclosure provides many different embodiments, or examples, for implementing different features of the invention. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

Further, spatially relative terms, such as "underlying," "below," "lower," "overlying," "upper" and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 80 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

An integrated package including a System on Integrate Chip (SoIC) package bonded to an Integrated Fan-Out (InFO) package and the method of forming the same are provided in accordance with various embodiments. The intermediate stages of forming the packages are illustrated in accordance with some embodiments. Some variations of some embodiments are discussed. Throughout the various views and illustrative embodiments, like reference numbers are used to designate like elements.

FIGS. 1 through 12 illustrate the cross-sectional views of intermediate stages in the formation of a SoIC package in accordance with some embodiments of the present disclosure. The steps shown in FIGS. 1 through 12 are also reflected schematically in the process flow 200 shown in FIG. 17.

FIG. 1 illustrates the cross-sectional view in the formation of wafer 2. In accordance with some embodiments of the present disclosure, wafer 2 is an interposer wafer, which is free from any active devices such as transistors and/or diodes therein. In accordance with some embodiments of the present disclosure, interposer wafer 2 is also free from passive devices such as capacitors, inductors, resistors, or the like therein. Interposer wafer 2 may include a plurality of metal lines and vias therein, with some details of one of

interposer dies **4** illustrated schematically. Interposer dies **4** are alternatively referred to as interposers or chips herein-after. Interposer dies **4** are used for routing, as will be discussed in subsequent paragraphs.

Wafer **2** may include substrate **20** and the features over the top surface of substrate **20**. In accordance with some embodiments of the present disclosure, substrate **20** is a semiconductor substrate. The substrate **20** may be formed of crystalline silicon, crystalline germanium, crystalline silicon germanium, and/or a III-V compound semiconductor such as GaAsP, AlInAs, AlGaAs, GaInAs, GaInP, GaInAsP, or the like. Semiconductor substrate **20** may also be a bulk silicon substrate or a Silicon-On-Insulator (SOI) substrate. In accordance with some embodiments in which the substrate **20** is a semiconductor substrate, Shallow Trench Isolation (STI) regions (not shown) may be formed in substrate **20** to isolate the regions in substrate **20**. In accordance with alternative embodiments, STI regions are not formed in wafer **2** since wafer **2** does not have active devices, and hence does not need STI regions to isolate active regions from each other. Substrate **20** may also be a dielectric substrate, which may be formed of silicon oxide, for example. In accordance with some embodiments, through-vias (not shown) are formed to extend into semiconductor substrate **20**, wherein the through-vias are used to electrically inter-couple the features on opposite sides of substrate **20**. In accordance with alternative embodiments, no through-vias are formed extending into semiconductor substrate **20**.

Dielectric layer **24** may be formed over substrate **20**. In accordance with some embodiments of the present disclosure, dielectric layer **24** is an Inter-Layer Dielectric (ILD), which may be formed of silicon oxide, Phospho Silicate Glass (PSG), Boro Silicate Glass (BSG), Boron-Doped Phospho Silicate Glass (BPSG), Fluorine-Doped Silicate Glass (FSG), Tetra Ethyl Ortho Silicate (TEOS), or the like. Dielectric layer **24** may be formed using thermal oxidation, spin coating, Flowable Chemical Vapor Deposition (FCVD), Atomic Layer Deposition (ALD), Chemical Vapor Deposition (CVD), Plasma Enhanced Chemical Vapor Deposition (PECVD), Low Pressure Chemical Vapor Deposition (LPCVD), or the like.

Over dielectric layer **24** resides interconnect structure **26**. Interconnect structure **26** includes metal lines **28** and vias **30**, which are formed in dielectric layers **32**. Dielectric layers **32** are alternatively referred to as Inter-Metal Dielectric (IMD) layers hereinafter. In accordance with some embodiments of the present disclosure, dielectric layers **32** are formed of low-k dielectric materials having dielectric constants (k-values) lower than 3.8. For example, the k values of dielectric layers **32** may be lower about 3.0 or lower than about 2.5. Dielectric layers **32** may be formed of Black Diamond (a registered trademark of Applied Materials), a carbon-containing low-k dielectric material, Hydrogen Silsesquioxane (HSQ), Methylsilsesquioxane (MSQ), or the like. In accordance with alternative embodiments of the present disclosure, some or all of dielectric layers **32** are formed of non-low-k dielectric materials such as silicon oxide, silicon carbide (SiC), silicon carbo-nitride (SiCN), silicon oxy-carbo-nitride (SiOCN), or the like. In accordance with some embodiments of the present disclosure, the formation of dielectric layers **32** includes depositing a porogen-containing dielectric material, and then performing a curing process to drive out the porogen, and hence the remaining dielectric layers **32** is porous. Etch stop layers (not shown), which may be formed of silicon carbide, silicon nitride, or the like, are formed between IMD layers **32**, and are not shown for simplicity.

Metal lines **28** and vias **30** are formed in dielectric layers **32**. The metal lines **28** at a same level are collectively referred to as a metal layer hereinafter. In accordance with some embodiments of the present disclosure, interconnect structure **26** includes a plurality of metal layers that are interconnected through vias **30**. Metal lines **28** and vias **30** may be formed of copper or copper alloys, and they can also be formed of other metals. The formation process may include single damascene and dual damascene processes. In a single damascene process, a trench is first formed in one of dielectric layers **32**, followed by filling the trench with a conductive material. A planarization process such as a CMP process is then performed to remove the excess portions of the conductive material higher than the top surface of the IMD layer, leaving a metal line in the trench. In a dual damascene process, both a trench and a via opening are formed in an IMD layer, with the via opening underlying and in spatial communication with the trench. The conductive material is then filled into the trench and the via opening to form a metal line and a via, respectively. The conductive material may include a diffusion barrier layer lining the trench and the via and a copper-containing metallic material over the diffusion barrier layer. The diffusion barrier layer may include titanium, titanium nitride, tantalum, tantalum nitride, or the like.

FIG. **1** illustrates surface dielectric layer **34** in accordance with some embodiments of the present disclosure. Surface dielectric layer **34** is formed of a non-low-k dielectric material such as silicon oxide. Surface dielectric layer **34** is alternatively referred to as a passivation layer since it has the function of isolating the underlying low-k dielectric layers (if any) from the adverse effect of detrimental chemicals and moisture. Surface dielectric layer **34** may also have a composite structure including more than one layer, which may be formed of silicon oxide, silicon nitride, Undoped Silicate Glass (USG), or the like. Interposer dies **4** may also include metal pads underlying surface dielectric layer **34**, and the metal pads may include aluminum or aluminum copper pads, Post-Passivation Interconnect (PPI), or the like, which are not shown for simplicity.

Bond pads **36A** and **36B**, which are also collectively and individually referred to bond pads **36**, are formed in surface dielectric layer **34**. In accordance with some embodiments of the present disclosure, bond pads **36A** and **36B** are formed through a single damascene process, and may also include barrier layers and a copper-containing material formed over the barrier layers. In accordance with alternative embodiments of the present disclosure, bond pads **36A** and **36B** are formed through a dual damascene process. Some bond pads **36A** may be electrically coupled to other bond pads **36A** and **36B** through metal lines **28** and vias **30**. In accordance with some embodiments of the present disclosure, each of bond pads **36A** and bond pads **36B** is electrically connected to at least one (or more) of other bond pads **36A** and **36B** through metal lines **28** and vias **30**, and none of bond pads **36A** and **36B** is electrically disconnected to all other bond pads **36A** and **36B**.

In accordance with some embodiments of the present disclosure, there is no organic dielectric material such as polymer, resin, and molding compound in wafer **2**. Organic dielectric layers typically have high Coefficients of Thermal Expansion (CTEs), such as 10 ppm/C° or higher. This is significantly greater than the CTE of silicon substrate (such as substrate **20**), which is about 3 ppm/C°. Accordingly, organic dielectric layers tend to cause the warpage of wafer **2**. Not including organic materials in wafer **2** advantageously reduces the CTE mismatch between the layers in

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wafer 2, and results in the reduction in the warpage of the resulting SoIC package 86 (FIG. 12). Also, not including organic materials in wafer 2 makes the formation of fine-pitch metal lines (such as 66 in FIG. 12) and high-density bond pads possible, and results in the improvement in the routing ability. The top surface dielectric layer 34 and bond pads 36 are planarized so that their top surfaces are coplanar, which may be resulted due to the CMP in the formation of bond pads 36.

Next, device dies 42A and 42B are bonded to wafer 2, as shown in FIG. 2. The respective process is illustrated as process 202 in the process flow shown in FIG. 22. In accordance with some embodiments of the present disclosure, device dies 42A and 42B are memory dies such as Dynamic Random Access Memory (DRAM) dies or Static Random Access Memory (SRAM) dies. Each of device dies 42A and 42B may also be a Central Processing Unit (CPU) die, a Micro Control Unit (MCU) die, an input-output (IO) die, a BaseBand (BB) die, or an Application processor (AP) die. Device dies 42A and 42B may be the same type or different types of dies selected from the above-listed types. Furthermore, device dies 42A and 42B may be formed using different technologies such as 45 nm technology, 28 nm technology, 20 nm technology, or the like. Dies 4, device die 42A and device dies 42B in combination function as a package, which may be a memory package or logic package.

Device dies 42A and 42B include substrates 44A and 44B, respectively, which may be semiconductor substrates such as silicon substrates. In accordance with some embodiments, the substrates 44A and 44B are also referred to as semiconductor substrates 44A and 44B. In accordance with some embodiments of the present disclosure, device dies 42A and 42B are free from Through-Silicon Vias (TSVs) therein. Also, device dies 42A and 42B include interconnect structures 48A and 48B, respectively, for connecting to the active devices and passive devices in device dies 42A and 42B. Interconnect structures 48A and 48B include metal lines and vias, which are illustrated schematically. Substrates 44A and 44B are free from through-vias therein. Accordingly, all external electrical connections of device dies 42A and 42B are made through bond pads 50A and 50B.

Device die 42A includes bond pads 50A and dielectric layer 52A at the illustrated bottom surface. The bottom surfaces of bond pads 50A are coplanar with the bottom surface of dielectric layer 52A. Device die 42B includes bond pads 50B and dielectric layer 52B at the illustrated bottom surface. The bottom surfaces of bond pads 50B are coplanar with the bottom surface of dielectric layer 52B. In accordance with some embodiments of the present disclosure, device dies 42A and 42B are free from organic dielectric materials such as polymer, resin, molding compound, and the like.

The bonding of device dies 42A and 42B to wafer 2 may be achieved through hybrid bonding. For example, bond pads 50A and 50B are bonded to bond pads 36A through metal-to-metal direct bonding. In accordance with some embodiments of the present disclosure, the metal-to-metal direct bonding is copper-to-copper direct bonding. Furthermore, dielectric layers 52A and 52B are bonded to surface dielectric layer 34, for example, with fusion bonds (which may include Si—O—Si bonds) generated.

To achieve the hybrid bonding, device dies 42A and 42B are first pre-bonded to surface dielectric layer 34 and bond pads 36A by lightly pressing device dies 42A and 42B against interposer die 4. Although two device dies 42A and 42B are illustrated, the hybrid bonding may be performed at wafer level, and a plurality of device die groups identical to

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the illustrated die group including device dies 42A and 42B is pre-bonded, and arranged as rows and columns.

After all device dies 42A and 42B are pre-bonded, an anneal is performed to cause the inter-diffusion of the metals in bond pads 36A and the corresponding overlying bond pads 50A and 50B. The annealing temperature may be in the range between about 200° and about 400° C., and may be in the range between about 300° and about 400° C. in accordance with some embodiments. The annealing time is in the range between about 1.5 hours and about 3.0 hours, and may be in the range between about 1.5 hours and about 2.5 hours in accordance with some embodiments. Through the hybrid bonding, bond pads 50A and 50B are bonded to the corresponding bond pads 36A through direct metal bonding caused by metal inter-diffusion.

Surface dielectric layer 34 is also bonded to dielectric layers 52A and 52B, with bonds formed therebetween. For example, the atoms (such as oxygen atoms) in one of surface dielectric layer 34 and dielectric layers 52A/52B form chemical or covalence bonds with the atoms (such as silicon atoms) in the other one of surface dielectric layers 34 and dielectric layer 52A/52B. The resulting bonds between surface dielectric layers 34 and dielectric layer 52A/52B are dielectric-to-dielectric bonds. Bond pads 50A and 50B may have sizes greater than, equal to, or smaller than, the sizes of the respective bond pads 36A. Gaps 46 are left between neighboring device dies 42A and 42B.

Further referring to FIG. 2, a backside grinding may be performed to thin device dies 42A and 42B, for example, to a thickness between about 15 μm and about 30 μm . FIG. 2 schematically illustrates dashed lines 44A-BS1 and 44B-BS1, which are the back surfaces of device dies 42A and 42B, respectively before the backside grinding. 44A-BS2 and 44B-BS2 are the back surfaces of device dies 42A and 42B, respectively after the backside grinding. Through the thinning of device dies 42A and 42B, the aspect ratio of gaps 46 between neighboring device dies 42A and 42B is reduced. Otherwise, the gap-filling may be difficult due to the otherwise high aspect ratio of gaps 46. In accordance with other embodiments in which the aspect ratio of gaps 46 is not too high for gap filling, the backside grinding is skipped.

FIG. 3 illustrates the formation of gap-filling layers/regions 54 and 56. The respective process is illustrated as process 204 in the process flow shown in FIG. 22. In accordance with some embodiments of the present disclosure, the gap-filling layers includes dielectric layer 54, and dielectric layer 56 over and contacting etch stop layer 54. Dielectric layer 54 may be deposited using a conformal deposition method such as Atomic Layer Deposition (ALD) or Chemical Vapor Deposition (CVD). In accordance with some embodiments, dielectric layer 54 is also referred to as an etch stop layer or a dielectric liner. Dielectric layer 56 may be formed using a conformal deposition method, or a non-conformal deposition method such as High-Density Plasma Chemical Vapor Deposition (HDPCVD), Flowable Chemical Vapor Deposition (CVD), spin-on coating, or the like. In accordance with some embodiments of the present disclosure, the gap-filling layers are free from organic materials such as polymers, resin, molding compounds, and the like.

Etch stop layer 54 is formed of a dielectric material that has a good adhesion to the top surfaces and sidewalls of device dies 42A and 42B and the top surfaces of surface dielectric layer 34 and bond pads 36B. Etch stop layer 54 also extends on the top surfaces of device dies 42A and 42B. In accordance with some embodiments of the present dis-

closure, etch stop layer **54** is formed of a nitride-containing material such as silicon nitride. The thickness **T1**, (including **T1A** and **T1B**) of etch stop layer **54** may be in the range between about 500 Å and about 1,000 Å. It is appreciated that the values recited throughout the description are examples, and different values may be used. Etch stop layer **54** may be a conformal layer, for example, with the thickness **T1A** of horizontal portions and thickness **T1B** of the vertical portions being substantially equal to each other, for example, with the difference (**T1A-T1B**) having an absolute value smaller than about 20 percent, or smaller than about 10 percent, of both thicknesses **T1A** and **T1B**.

Dielectric layer **56** is formed of a material different from the material of etch stop layer **54**. Dielectric layer **56** may be formed of an inorganic dielectric material. In accordance with some embodiments of the present disclosure, dielectric layer **56** includes an oxide such as silicon oxide, which may be formed of TEOS, while other dielectric materials such as silicon carbide, silicon oxynitride, silicon oxy-carbo-nitride, or the like may also be used when there is an adequate etching selectivity (for example, higher than about so) between dielectric layer **56** and etch stop layer **54**. The etching selectivity is the ratio of the etching rate of dielectric layer **56** to the etching rate of etch stop layer **54** when etching dielectric layer **56** in a subsequent process. Dielectric layer **56** fully fills gaps **46** (FIG. 2), and further includes some portions overlapping device dies **42A** and **42B**. Dielectric layer **56** may be formed of a non-conformal formation method or a conformal formation method.

A planarization process such as a CMP process or a mechanical grinding process is performed to remove the excess portions of dielectric layer **56**. In accordance with some embodiments of the present disclosure, the planarization is stopped when there is a layer of dielectric layer **56** overlapping device dies **42A** and **42B**. Accordingly, etch stop layer **54** is not polished. In accordance with alternative embodiments of the present disclosure, the planarization is performed using etch stop layer **54** as a CMP stop layer. As a result, when the planarization is stopped, the top surfaces **54A** of etch stop layer **54** are exposed, and there are remaining horizontal portions of etch stop layer **54** overlapping device dies **42A** and **42B**. In accordance with yet other embodiments of the present disclosure, the planarization is stopped after substrates **44A** of device die **42A** and substrate **44B** of device die **42B** are exposed. The remaining portions of etch stop layers **54** and dielectric layer **56** are collectively referred to as (gap-filling) isolation regions **58**. Isolation regions **58** are also referred to as inorganic gap-filling (or gap-fill) regions.

FIG. 4 illustrates the etching of dielectric layer **56** to form openings **59**. The respective process is illustrated as process **206** in the process flow shown in FIG. 22. In accordance with some embodiments of the present disclosure, a photo resist (not shown) is formed and patterned, and dielectric layer **56** is etched using the patterned photo resist as an etching mask. Openings **59** are thus formed, and extend down to etch stop layer **54**. In accordance with some embodiments of the present disclosure, dielectric layer **56** comprises an oxide, and the etching may be performed through dry etching. The etching gas may include a mixture of NF_3 and NH_3 , or a mixture of HF and NH_3 . Using etch top layer **54** to stop the etching for forming openings **59** allows the downward proceeding of multiple openings **59** on the same wafer **2** to be synchronized at the same intermediate level, so that the faster-etched openings **59** will wait for the slower-etched openings **59** before they extend downwardly again.

Next, etch stop layer **54** is etched, so that openings **59** extend down to bond pads **36B**. In accordance with some embodiments of the present disclosure, etch stop layer **54** comprises silicon nitride, and the etching is performed using dry etching. The etching gas may include a mixture of CF_4 , O_2 , and N_2 , a mixture of NF_3 and O_2 , SF_6 , or a mixture of SF_6 and O_2 .

FIG. 5 illustrates the formation of through-vias **60**, which fills openings **59** (FIG. 4), and are connected to bond pads **36B**. The respective process is illustrated as process **208** in the process flow shown in FIG. 22. In accordance with some embodiments of the present disclosure, the formation of through-vias **60** includes performing a plating process such as an electrical-chemical plating process or an electro-less plating process. Through-vias **60** may include a metallic material such as tungsten, aluminum, copper, or the like. A conductive barrier layer (such as titanium, titanium nitride, tantalum, tantalum nitride, or the like) may also be formed underlying the metallic material. A planarization process such as a CMP process is performed to remove excess portions of the plated metallic material, and the remaining portions of the metallic material form through-vias **60**. Through-vias **60** may have substantially straight and vertical side walls. Also, through-vias **60** may have a tapered profile, with top widths **WT** slightly greater than the respective bottom widths **WB**. In accordance with some embodiments, as shown in FIG. 5, a single through-via **60** is formed to contact each of bond pads **36B**. In accordance with alternative embodiments, a plurality of through-via **60** (such as two or three) are formed over and contacting a same bond pad **36B**.

Referring to FIG. 6, dielectric layer **62**, which may be an inorganic layer, is formed. The respective process is illustrated as process **210** in the process flow shown in FIG. 22. In accordance with some embodiments of the present disclosure, dielectric layer **62** is formed of a low-k dielectric material having a k value lower than 3.8, and the k value may be lower than about 3.0, and lower than about 2.5, for example. In accordance with alternative embodiments, dielectric layer **62** is formed of an oxide such as silicon oxide, a nitride such as silicon nitride, or the like. Dielectric layer **62** is then patterned in a lithography process to form openings **64**, and through-vias **60** are exposed.

Metal features **66** are then formed, as shown in FIG. 7. The respective process is also illustrated as process **210** in the process flow shown in FIG. 22. Metal features **66** may include metal lines and metal pads, and may be formed using a damascene process, which includes depositing a conformal conductive barrier layer into openings **64** (FIG. 6), plating a metallic material such as copper or a copper alloy, and performing a planarization to remove excess portions of metal features **66**. Metal features **66** may have a single damascene structure as illustrated in FIG. 7. In accordance with other embodiments of the present disclosure, metal features **66** have a dual damascene structure.

In accordance with some embodiments of the present disclosure, the combined structure including dielectric layer **62** and all underlying structures are free from organic materials (such as polymer layers, molding compound, resin, or the like), so that the process for forming metal features **66** may adopt the process used for forming device dies, and fine-pitches metal lines **66** having small pitches and line-widths are made possible.

FIG. 8 illustrates the formation of metal pads **68**. The respective process is illustrated as process **212** in the process flow shown in FIG. 22. In accordance with some embodiments, metal pads **68** are formed of aluminum copper. The

formation may include depositing a metal layer, and patterning the metal layer. The remaining portions of the etched metal layer are metal pads **68**.

FIG. **9** illustrates the formation of passivation layer **70** and polymer layer **72**. Passivation layer **70** is formed over and contacting dielectric layer **62** in accordance with some embodiments. Passivation layer **70** may be a single layer or a composite layer, and may be formed of a non-porous material. In accordance with some embodiments of the present disclosure, passivation layer **70** is a composite layer including a silicon oxide layer (not shown separately), and a silicon nitride layer (not shown separately) over the silicon oxide layer. Passivation layer **70** may also be formed of other non-porous dielectric materials such as Un-doped Silicate Glass (USG), silicon oxynitride, and/or the like.

Next, passivation layer **70** is patterned, so that some portions of metal pads **68** are exposed through openings in passivation layer **70**. Polymer layer **72** is then formed. The respective process is illustrated as process **214** in the process flow shown in FIG. **22**. Polymer layer **72** may be formed of polyimide, polybenzoxazole (PBO), or the like. Polymer layer **72** is also patterned to form openings, through which metal pads **68** are exposed. In accordance with some embodiments, polymer layer **72** has a great thickness, which may be in the range between about 3 μm and about 6 μm .

Referring to FIG. **10**, Redistribution Lines (RDLs) **74** are formed, and the via-ports of RDLs **74** extend into the openings in polymer layer **72** (FIG. **9**) to electrically connect to metal pads **68**. The respective process is illustrated as process **216** in the process flow shown in FIG. **22**. It is appreciated that RDLs **74** may include metal pads and metal lines, and may be used for routing, so that the metal pads in RDLs **74** may be rerouted into the regions overlapping device dies **42A** and **42B**.

FIG. **11** illustrates the formation of polymer layer **76**, which may be formed of polyimide, PBO, or the like. The respective process is illustrated as process **218** in the process flow shown in FIG. **22**. Openings **78** are formed in polymer layer **76** to reveal RDLs **74**. In accordance with some embodiments, polymer layer **76** has a great thickness, which may be in the range between about 5 μm and about 10 μm . Since polymer layers **72** and **76** have low Young's moduli, which are much lower than the Young's moduli in underlying layers formed of inorganic material, polymer layers **72** and **76** can absorb the stress in the resulting package. With polymer layers **72** and **76** having great thicknesses, their ability for absorbing stress is improved. For example, the structure underlying polymer layers **72** and **76**, which underlying structure includes wafer **2**, device dies **42A** and **42B**, and dielectric regions **56**, may or may not include polymer. When the underlying layers do not include polymers, the package may benefit from polymer layers **72** and **76** for their ability for absorbing stress.

Referring to **12**, Under-bump metallurgies (UBMs) **80** are formed, and UBMs **80** extend into polymer layer **76** to connect to RDLs **74**. In accordance with some embodiments of the present disclosure, each of UBMs **80** includes a barrier layer (not shown) and a seed layer (not shown) over the barrier layer. The barrier layer may be a titanium layer, a titanium nitride layer, a tantalum layer, a tantalum nitride layer, or a layer formed of a titanium alloy or a tantalum alloy. The materials of the seed layer may include copper or a copper alloy. Other metals such as silver, gold, aluminum, palladium, nickel, nickel alloys, tungsten alloys, chromium, chromium alloys, and combinations thereof may also be included in UBMs **80**. In accordance with some embodiments, the formation of UBMs **80** include depositing a

blanket barrier layer and a blanket seed layer, forming a patterned etching mask (such as a patterned photo resist) over the seed layer, and then etching the blanket seed layer and the blanket barrier layer. In accordance with other embodiments, the formation of UBMs **80** include depositing a blanket barrier layer and a blanket seed layer, forming a patterned plating mask (such as a patterned photo resist) over the blanket seed layer, plating metal pillars in the openings in the patterned plating mask, removing the patterned plating mask, and then etching the portions of the blanket seed layer and the blanket barrier layer previously covered by the patterned plating mask.

As also shown in FIG. **12**, electrical connectors **82** are formed over and contacting UBMs **80**. The respective process is illustrated as process **220** in the process flow shown in FIG. **22**. Electrical connectors **82** may include metal pillars, solder regions, or the like. Throughout the description, the structure shown in FIG. **12** is referred to as composite wafer **84**. A die-saw (singulation) step is performed on composite wafer **84** to separate composite wafer **84** into a plurality of SoIC packages **86**. The respective process is illustrated as process **222** in the process flow shown in FIG. **22**.

FIGS. **13** through **18** illustrate the cross-sectional views of intermediate stages in the formation of an Integrated Fan-Out (InFO) package in accordance with some embodiments. Referring to FIG. **13**, carrier **120** is provided, and release film **122** is formed on carrier **120**. Carrier **120** is formed of a transparent material, and may be a glass carrier, a ceramic carrier, an organic carrier, or the like. Release film **122** may be formed of a Light-To-Heat-Conversion (LTHC) coating material, which is applied onto carrier **120** through coating. In accordance with some embodiments, the release film **122** is also referred to as a LTHC coating material. In accordance with some embodiments of the present disclosure, the LTHC coating material is capable of being decomposed under the heat of light/radiation (such as laser), and hence can release carrier **120** from the structure formed thereon.

In accordance with some embodiments, as also shown in FIG. **1**, polymer buffer layer **124** is formed on LTHC coating material **122**. In accordance with some embodiments, polymer buffer layer **124** is formed of PBO, polyimide, benzocyclobutene (BCB), or another applicable polymer.

Metal seed layer **126** is formed, for example, through Physical Vapor Deposition (PVD). The respective process is illustrated as process **230** in the process flow shown in FIG. **22**. Metal seed layer **126** may be in physical contact with polymer buffer layer **124**. In accordance with some embodiments of the present disclosure, metal seed layer **126** includes a titanium layer and a copper layer over the titanium layer. In accordance with alternative embodiments of the present disclosure, metal seed layer **126** includes a copper layer contacting LTHC coating material **122**.

Photo resist **128** is formed over metal seed layer **126**. The respective process is also illustrated as process **230** in the process flow shown in FIG. **22**. A light-exposure is then performed on photo resist **128** using a photo lithography mask (not shown). After a subsequent development, openings **130** are formed in photo resist **128**. Some portions of metal seed layer **126** are exposed through openings **130**. Next, metal posts **132** are formed by plating a metallic material in openings **130**. The plated metallic material may be copper or a copper alloy. The respective process is illustrated as process **232** in the process flow shown in FIG. **22**.

In subsequent steps, photo resist **128** is removed, and hence the underlying portions of metal seed layer **126** are

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exposed. The exposed portions of metal seed layer **126** are then removed in an etching step, for example, in an anisotropic or an isotropic etching step. The edges of the remaining metal seed layer **126** are thus co-terminus with the respective overlying portions of metal posts **132**. The resulting metal posts **132** are illustrated in FIG. **14**. Throughout the description, the remaining portions of metal seed layer **126** are considered as parts of metal posts **132**, and may not be illustrated separately. The top-view shapes of metal posts **132** include, and are not limited to, circular shapes, rectangles, hexagons, octagons, and the like.

FIG. **15** illustrates the placement/attachment of device die **136**. The respective process is illustrated as process **234** in the process flow shown in FIG. **22**. Device die **136** is attached to polymer buffer layer **124** through Die-Attach Film (DAF) **138**, which is an adhesive film pre-attached on device die **136** before device die **136** is placed on polymer buffer layer **124**. Accordingly, DAF **138** and device die **136**, before attached to polymer buffer layer **124**, are in combination an integrated piece. Device die **136** may include a semiconductor substrate having a back surface (the surface facing down) in physical contact with DAF **138**. Device die **136** may include integrated circuit devices (such as active devices, which include transistors, for example, not shown) at the front surface (the surface facing up) of the semiconductor substrate. In accordance with some embodiments of the present disclosure, device die **136** is a logic die, which may be a Central Processing Unit (CPU) die, a Graphic Processing Unit (GPU) die, a mobile application die, a Micro Control Unit (MCU) die, an input-output (IO) die, a BaseBand (BB) die, an Application processor (AP) die, or the like. Since carrier **120** is at wafer level, although one device die **136** is illustrated, a plurality of device dies **136** is placed over polymer buffer layer **124**, and may be allocated as an array including a plurality of rows and a plurality of columns.

In accordance with some exemplary embodiments, metal pillars **142** (such as copper pillars) are pre-formed as portions of device die **136**, and metal pillars **142** are electrically coupled to the integrated circuit devices such as transistors (not shown) in device die **136**. In accordance with some embodiments of the present disclosure, a dielectric material such as a polymer fills the gaps between neighboring metal pillars **142** to form top dielectric layer **144**. Top dielectric layer **144** may also include a portion covering and protecting metal pillars **142**. Top dielectric layer **144** may be a polymer layer, which may be formed of PBO or polyimide in accordance with some embodiments of the present disclosure.

Next, device die **136** and metal posts **132** are encapsulated in encapsulating material **148**, as shown in FIG. **16**. The respective process is illustrated as process **236** in the process flow shown in FIG. **22**. Encapsulating material **148** may include a molding compound, a molding underfill, an epoxy, and/or a resin. When formed of molding compound, encapsulating material **148** may include a base material, which may be a polymer, a resin, an epoxy, or the like, and filler particles (not shown) in the base material. The filler particles may be dielectric particles of SiO₂, Al₂O₃, silica, or the like, and may have spherical shapes. Also, the spherical filler particles may have a plurality of different diameters. Both the filler particles and the base material in the molding compound may be in physical contact with polymer buffer layer **124**.

The top surface of encapsulating material **148**, as disposed, is higher than the top ends of metal pillars **142** and metal posts **132**. In a subsequent step, as shown in FIG. **16**,

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a planarization process such as a CMP process or a mechanical grinding process is performed to thin encapsulating material **148** and top dielectric layer **144**, until metal posts **132** and metal pillars **142** are exposed. Metal posts **132** are alternatively referred to as through-vias **132** since they penetrate through encapsulating material **148**. Due to the planarization process, the top ends of through-vias **132** are substantially level (coplanar) with the top surfaces of metal pillars **142**, and are substantially coplanar with the top surface of encapsulating material **148**.

FIG. **17** illustrates the formation of a front-side redistribution structure iso, which includes dielectric layers **152** and RDLs **154** in dielectric layers **152**. The respective process is illustrated as process **238** in the process flow shown in FIG. **22**. In accordance with some embodiments of the present disclosure, dielectric layers **152** are formed of polymers such as PBO, polyimide, or the like. In accordance with alternative embodiments of the present disclosure, dielectric layers **152** are formed of inorganic dielectric materials such as silicon nitride, silicon oxide, or the like.

In accordance with some embodiments of the present disclosure, the formation of a dielectric layer **152** and the corresponding RDLs **154** may include depositing dielectric layer **152**, forming via openings in the corresponding dielectric layer **152** to expose the underlying conductive features, depositing a metal seed layer (not shown), forming and patterning a photo resist (not shown) over the corresponding metal seed layer, and plating a metallic material such as copper and/or aluminum over the metal seed layer. The patterned photo resist is then removed, followed by etching the portions of the metal seed layer previously covered by the patterned photo resist.

FIG. **18** illustrates the formation of UBMs **156**. The top dielectric layer **152** is patterned to form openings, and UBMs **156** are formed to extend into the openings in the top dielectric layer **152** to contact the metal pads in RDLs **154**. UBMs **156** may be formed of nickel, copper, titanium, or multi-layers thereof. In accordance with some exemplary embodiments, UBMs **156** include a titanium layer and a copper layer over the titanium layer.

Electrical connectors **158** are then formed. The respective process is illustrated as process **240** in the process flow shown in FIG. **22**. The formation of electrical connectors **158** may include placing solder balls on the exposed portions of UBMs **156**, and then reflowing the solder balls into solder regions. Throughout the description, the structure including polymer buffer layer **124** and the overlying structure in combination is referred to as package **160**, which is a composite wafer (and also referred to as composite wafer **160** hereinafter) including a plurality of device dies **136**. Next, composite wafer **160** is demounted from carrier **120**, for example, by projecting a laser beam onto LTHC coating material **122**, so that LTHC coating material **122** is decomposed, releasing composite wafer **160** from carrier **120**. Composite wafer **160** includes a plurality of InFO packages **162** therein.

FIG. **19** illustrates the formation of package **88**, which includes bonding SoIC package **86** to InFO package **162**. The respective process is illustrated as process **242** in the process flow shown in FIG. **22**. In accordance with some embodiments of the present disclosure, a plurality of SoIC packages **86** are bonded to composite wafer **160**, with electrical connectors **82** of SoIC packages **86** penetrating through polymer buffer layer **124** of InFO package **162** to bond to through-vias **132**. Underfill regions **90** are dispensed into the gaps between SoIC packages **86** and InFO packages **162**. The resulting composite wafer **160** is then singulated,

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resulting in a plurality of packages 88. The respective process is illustrated as process 244 in the process flow shown in FIG. 22.

FIGS. 20 and 21 illustrate packages 88 in accordance with some embodiments of the present disclosure. The packages 88 in accordance with these embodiments are similar to package 88 as shown in FIG. 19, except some features in FIG. 19 are omitted. In accordance with some embodiments of the present disclosure, the metal pads 68 and passivation layer 70 as shown in FIG. 19 may be omitted. The resulting figure is illustrated in FIG. 20. In accordance with some other embodiments of the present disclosure, the metal pads 68, passivation layer 70, RDLs 74, and polymer layer 72 as shown in FIG. 19 are omitted. The resulting figure is illustrated in FIG. 21. In accordance with some embodiments of the present disclosure, when polymer layer 72 is omitted, the thickness of polymer layer 76 may (or may not) be increased, for example, to about 8 μm and about 16 μm , so that the polymer's ability for absorbing stress is maintained.

In above-illustrated embodiments, some processes and features are discussed in accordance with some embodiments of the present disclosure. Other features and processes may also be included. For example, testing structures may be included to aid in the verification testing of the 3D packaging or 3DIC devices. The testing structures may include, for example, test pads formed in a redistribution layer or on a substrate that allows the testing of the 3D packaging or 3DIC, the use of probes and/or probe cards, and the like. The verification testing may be performed on intermediate structures as well as the final structure. Additionally, the structures and methods disclosed herein may be used in conjunction with testing methodologies that incorporate intermediate verification of known good dies to increase the yield and decrease costs.

The embodiments of the present disclosure have some advantageous features. Since the portions of SoIC package (such as what is shown in FIG. 8) are free from polymers, resins, and molding compounds, the CTE mismatch in this part of SoIC package is reduced, and the stress inside this part of the SoIC package is reduced. It is thus possible to form fine-pitch RDLs due to the reduction in the warpage. Some inorganic materials are hard and have high Young's moduli. The SoIC packages, if free from polymers, will have high hardness values. For example, silicon nitride, which is used in the SoIC package, has a Young's modulus greater than loft Polymers such as polyimide and PBO, on the other hand, have Young's moduli equal to about 3.5 or lower. Accordingly, the added polymer layers may absorb the stress that cannot be absorbed by the hard inorganic materials in the SoIC packages. Experiment results indicate that if SoIC packages are free from polymers, after the SoIC packages are bonded to the composite wafer including the un-sawed InFO packages, the bonding may break, and the SoIC packages will fall off from the composite wafer. By forming polymer layers to absorb the stress, the bonding between the SoIC packages and the composite wafer is not damaged by the stress.

In accordance with some embodiments of the present disclosure, a method includes forming a first package comprising: bonding a first device die to an interposer wafer, wherein the interposer wafer comprises metal lines and vias; forming a gap-fill region to encircle the first device die; forming a through-via to penetrate through the gap-fill region, wherein the through-via is electrically connected to the first device die through the metal lines and the vias in the interposer wafer; forming a polymer layer over the gap-fill

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region; forming an electrical connector, wherein the electrical connector is electrically coupled to the through-via through a conductive feature in the polymer layer; and sawing the interposer wafer to separate the first package from other packages. In an embodiment, the forming the gap-fill region comprises forming a dielectric liner on surfaces of the interposer wafer and the first device die; filling a dielectric material on the dielectric liner; and planarizing the dielectric material. In an embodiment, after the planarizing, the dielectric liner comprises a portion overlapping the first device die. In an embodiment, the interposer wafer is free from active devices therein. In an embodiment, all dielectric materials underlying the polymer layer are inorganic materials. In an embodiment, the method further includes forming a second package comprising: forming a metal post; and encapsulating the metal post and a second device die in an encapsulating material; and bonding the second package to the first package. In an embodiment, the method further includes bonding a second device die to the interposer wafer, wherein the metal lines and vias electrically connect the first device die to the second device die, and the gap-fill region comprises a portion separating the first device die from the second device die. In an embodiment, the gap-filling region is formed of inorganic dielectric materials.

In accordance with some embodiments of the present disclosure, a method includes forming a first package comprising bonding a first device die and a second device die to an interposer die; encapsulating the first device die and the second device die in inorganic gap-fill materials; forming through-vias on metal pads of the interposer die, wherein the through-vias penetrate through the inorganic gap-fill materials, and are electrically connect to the first device die and the second device die through the interposer die; forming a dielectric layer over the first device die, the second device die, and the through-vias; forming metal features in the dielectric layer, wherein the metal features are formed using a damascene process; forming a polymer layer over the metal features, wherein all dielectric materials underlying the polymer layer are inorganic materials; and forming an electrical connector over the polymer layer; and bonding the first package to a second package, wherein the electrical connector is bonded to the second package. In an embodiment, the encapsulating the first device die and the second device die comprises depositing an etch stop layer contacting the first device die, the second device die, and the interposer die; forming a dielectric material over the etch stop layer; and planarizing the dielectric material. In an embodiment, the interposer die is free from active devices and passive devices, and all electrical connections from the first device die and the second device die to the second package are through the interposer die. In an embodiment, the forming the dielectric layer comprises forming a first low-k dielectric layer. In an embodiment, the polymer layer is in physical contact with the first low-k dielectric layer. In an embodiment, the interposer die comprises a second low-k dielectric layer, and the first low-k dielectric layer and the second low-k dielectric layer are on opposite sides of the inorganic gap-fill materials.

In accordance with some embodiments of the present disclosure, a device includes a first package comprising an interposer die free from active devices therein; a first device die and a second device die bonded to the interposer die; an inorganic gap-fill region encapsulating the first device die and the second device die therein; first through-vias penetrating through the inorganic gap-fill region, wherein the first through-vias are electrically connected to the first

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device die and the second device die through the interposer die; a dielectric layer over the first device die, the second device die, and the first through-vias; a polymer layer over the dielectric layer, wherein all dielectric materials underlying the polymer layer are inorganic materials; and an electrical connector over the polymer layer. In an embodiment, the device further includes a second package over and bonded to the first package, wherein the second package comprises: a third device die; a molding compound encapsulating the third device die therein; and second through-vias penetrating through the molding compound, wherein one of the second through-vias is bonded to the electrical connector. In an embodiment, the device further includes metal features in the dielectric layer, wherein the metal features have a damascene structure. In an embodiment, the dielectric layer is a low-k dielectric layer. In an embodiment, the inorganic gap-fill region comprises a silicon nitride layer lining surfaces of the interposer die, the first device die, and the second device die; and an oxide layer over the silicon nitride layer, wherein both the silicon nitride layer and the oxide layer comprise a portion overlapping the first device die. In an embodiment, the first device die and the second device die are electrically interconnected through conductive lines in the interposer die.

The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

What is claimed is:

1. A structure comprising:
 - a first package comprising:
 - an interposer die;
 - an inorganic gap-fill region over the interposer die;
 - a device die over and bonding to the interposer die, wherein the device die is in the inorganic gap-fill region;
 - an inorganic dielectric layer over the device die and the inorganic gap-fill region;
 - an organic polymer layer over the inorganic dielectric layer, wherein all dielectric materials that are underlying the organic polymer layer and also in the first package are inorganic materials; and
 - an electrical connector over the organic polymer layer and electrically coupling to the interposer die.
2. The structure of claim 1 further comprising a metal pad over the inorganic gap-fill region and electrically connecting to the interposer die, wherein both of the inorganic dielectric layer and the organic polymer layer physically contact a top surface of the metal pad.
3. The structure of claim 2, wherein the organic polymer layer comprises:
 - a first portion over the inorganic dielectric layer; and
 - a second portion in the inorganic dielectric layer.
4. The structure of claim 1, wherein the inorganic gap-fill region comprises:
 - an etch stop layer comprising a first portion over and contacting the device die; and

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a gap-filling layer over and contacting the etch stop layer, wherein the gap-filling layer comprises a second portion overlapping the first portion of the etch stop layer.

5. The structure of claim 4, wherein the etch stop layer forms an interface with a top surface of the device die.

6. The structure of claim 1 further comprising a second package over and bonded to the first package, wherein the second package comprises:

- a second device die;
- a molding compound encapsulating the second device die therein; and
- a through-via penetrating through the molding compound, wherein the through-via is bonded to the electrical connector.

7. The structure of claim 1, wherein entirety of the interposer die and the device die are free from organic dielectric materials.

8. The structure of claim 1, wherein a first dielectric layer of the device die is bonded to a second dielectric layer of the interposer die, and a first bond pad of the device die is bonded to a second bond pad of the interposer die.

9. The structure of claim 1, wherein the inorganic dielectric layer comprises an inorganic low-k dielectric material.

10. The structure of claim 1, wherein the interposer die comprises:

- a plurality of dielectric layers; and
- a plurality of metal layers in the plurality of dielectric layers, wherein first top surfaces of the plurality of metal layers are coplanar with second top surfaces of respective ones of the plurality of dielectric layers.

11. The structure of claim 1, wherein the interposer die comprises a semiconductor substrate, and the interposer die is free from through-vias penetrating through the semiconductor substrate.

12. The structure of claim 1 further comprising a through-via penetrating through the inorganic gap-fill region and electrically connecting to the interposer die.

13. A structure comprising:

- a first package comprising:
 - a device die; and
 - inorganic gap-fill layers encapsulating the device die therein;
- an inorganic dielectric layer overlapping both of the device die and the inorganic gap-fill layers;
- an organic polymer layer over and contacting the inorganic dielectric layer;
- a second package over and bonding to the first package; and
- an underfill between and contacting the second package and the organic polymer layer.

14. The structure of claim 13, wherein the inorganic gap-fill layers comprise:

- an etch stop layer comprising a first portion over and contacting a top surface of the device die, and a second portion contacting a sidewall of the device die; and
- an additional gap-filling layer over and contacting the first portion of the etch stop layer.

15. The structure of claim 14, wherein the first portion of the etch stop layer is spaced apart from the inorganic dielectric layer by the additional gap-filling layer.

16. The structure of claim 14, wherein the device die comprises a semiconductor substrate in physical contact with the first portion of the etch stop layer.

17. The structure of claim 13, wherein all dielectric layers that are in the first package and underlying the organic polymer layer are inorganic dielectric layers.

- 18.** A structure comprising:
a first package comprising:
a first interconnect structure comprising a plurality of
dielectric layers and a plurality of metal lines in the
plurality of dielectric layers; 5
a device die over and bonding to the first interconnect
structure;
an inorganic dielectric region comprising an etch stop
layer and a dielectric filling layer over the etch stop
layer, wherein both of the etch stop layer and the 10
dielectric filling layer comprise:
a first portion encircling the device die; and
a second portion overlapping the device die;
a second interconnect structure over the device die and
the inorganic dielectric region, wherein the second 15
interconnect structure comprises an organic polymer
layer; and
a solder region over and electrically connected to the
second interconnect structure;
a second package bonded to the first package through the 20
solder region; and
an underfill in physical contact with both of the organic
polymer layer and the second package.
- 19.** The structure of claim **18**, wherein the etch stop layer
comprises a portion separating the device die from the 25
organic polymer layer.
- 20.** The structure of claim **18**, wherein the second inter-
connect structure further comprises an inorganic low-k
dielectric layer between the organic polymer layer and the
second portion of the inorganic dielectric region. 30

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